

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

FORESIGHT DIAGNOSTICS, INC.,
Petitioner,

v.

PERSONALIS, INC.,
Patent Owner.

IPR2023-00224
Patent 11,384,394 B2

Before ULRIKE W. JENKS, ROBERT A. POLLOCK, and
TIMOTHY G. MAJORS, *Administrative Patent Judges*.

POLLOCK, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review challenging the patentability of claims 1–19 of U.S. Patent No. 11,384,394 B2 (“the ’394 patent,” Ex. 1001). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e) (2018). Having reviewed the parties’ arguments and cited evidence, for the reasons discussed below, we find that Petitioner has demonstrated by a preponderance of the evidence that claims 1–19 are unpatentable for the reasons set forth in the Petition.

A. Procedural History

Foresight Diagnostics Inc. (“Petitioner” or “Foresight”) filed a Petition for an *inter partes* review of claims 1–19 of the ’394 patent. Paper 1 (“Pet.”). Personalis, Inc. (“Patent Owner” or “Personalis”) timely filed a Preliminary Response. Paper 7. With our authorization (*see* Ex. 3001), Petitioner filed a Reply to the Preliminary Response (Paper 8); Patent Owner filed a corresponding Sur-reply (Paper 10). In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 11 (“Institution Decision” or “DI”).

After institution, Patent Owner filed a Response. Paper 18 (“POR”). Petitioner filed a (corrected) Reply. Paper 35 (“Reply”). Patent Owner filed a Sur-reply. Paper 40 (“Sur-reply”). With our authorization, Petitioner further filed a Response to Patent Owner’s Sur-reply. Paper 44 (“Resp.”).

On March 19, 2024, we held an oral hearing, the transcript of which is of record. Paper 48 (“Tr.”).

B. Real Parties-in-Interest

Petitioner identifies itself, Foresight Diagnostics Inc., as the real party-in-interest. Pet. 5. Patent Owner also identifies only itself, Personalis, Inc., as the real party-in-interest in this proceeding. Paper 6, 1.

C. Related Matters

In addition to the '394 patent at issue here, Petitioner concurrently challenges claims of related U.S. Patent No. 11,408,033 ("the '033 patent") in IPR2023-00317.

Patent Owner further identifies as "related proceedings," Petitioner's challenge of Personalis's U.S. Patent Nos. 11,299,783 ("the '783 patent") and 10,450,611 ("the '611 patent") in IPR2023-00545 and IPR2023-00546, respectively. Paper 6, 1. IPR2024-00170 involving U.S. Patent No. 11,584,968 ("the '968 patent") is also before us.

The '394, '033, '738, and '611 patents are at issue in *Personalis, Inc. v. Foresight Diagnostics, Inc.*, C.A. No. 1:22-cv-01913 (D. Colo.) ("the 01913 litigation"). See Pet. 5; Paper 6, 1; Ex. 1010, 1, 3–6; IPR2023-00317, Exhibit 1010, 1 (First Amended Complaint adding '033 patent). Petitioner further identifies as a "related matter[]," *Personalis, Inc. v. Foresight Inc.*, 1:23-cv-01623 (D. Colo.), which further involves the '968 patent. Paper 33, 2; Ex. 1010, 1; IPR2024-00170, Paper 1, 1, 3 (stating that Foresight has moved to consolidate the two district court proceedings).

D. Asserted Challenges to Patentability

Petitioner challenges the patentability of claims 1–19 on the following bases: (Pet. 18):

Ground	Claims Challenged	35 U.S.C. §	Reference(s)/Basis
1	1–11, 13–19	§ 103(a) ¹	Leary ²
2	1–11, 13–19	§ 103(a)	Leary, Ley ³
3	1–19	§ 103(a)	Leary, Liao, ⁴ Chan ⁵

Petitioner’s challenges are based on Leary (Ground 1), Leary in combination with Ley (Ground 2), and Leary in combination with Liao and Chan (Ground 3). Of the four asserted references, Patent Owner challenges only the prior art status of Chan, which we address in Section II.F.4, below.

In support of its patentability challenge, Petitioner relies on, *inter alia*, the First and Second Declarations of John Quackenbush, Ph.D., Exhibits 1020 and 1225, respectively. Patent Owner relies on, *inter alia*, the

¹ The Leahy-Smith America Invents Act (“AIA”) included revisions to 35 U.S.C. §§ 102 and 103 that became effective on March 16, 2013. Because the ’394 patent issued from an application that claims benefit of priority of an application filed before March 16, 2013, we apply the pre-AIA versions of the statutory bases for unpatentability.

² Leary *et al.*, “Development of Personalized Tumor Biomarkers Using Massively Parallel Sequencing,” 2(20) *Science Translational Medicine* 20ra14 (2010). Ex. 1002; Ex. 1026 (supplemental materials available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2858564/>).

³ Ley *et al.*, “DNA sequencing of a cytogenetically normal acute myeloid leukaemia genome,” 456 *Nature* 66–72 (2008). Ex. 1012.

⁴ Liao *et al.*, “Targeted Massively Parallel Sequencing of Maternal Plasma DNA Permits Efficient and Unbiased Detection of Fetal Alleles,” 57(1) *Clin. Chem.* 92–101 (2011). Ex. 1009.

⁵ Chan *et al.*, “Cancer Genome Scanning in Plasma: Detection of Tumor-Associated Copy Number Aberrations, Single-Nucleotide Variants, and Tumoral Heterogeneity by Massively Parallel Sequencing,” 59(1) *Clin. Chem.* 211–224 (2013). Ex. 1008.

First and Second Declarations of Henry Morrice Furneaux, Ph.D., Exhibits 2007 and 2031, respectively. Patent Owner further relies on the Declarations of Jonathan MacQuitty (Ex. 2033), Doug Zeman (Ex. 2034), Dan Norton (Ex. 2035), and John West (Ex. 2032).⁶

E. The '394 Patent and Related Background

1. Priority

The '394 patent, titled Methods and Systems for Genetic Analysis, issued to Bartha et al., from U.S. Application 17/548,379, filed December 10, 2021, via a series of continuation and division applications first filed on December 27, 2013, and further claims benefit of priority to U.S. Provisional Application No. 61/753,828, filed on January 17, 2013 (“the '828 Provisional Application”). Ex. 1001, code (12), (21), (22), (54), and (60). The parties dispute whether the '394 patent is entitled to the benefit of the provisional filing date. *See, e.g.*, Reply 25; POR 39.

2. Abstract and Specification

The Abstract of the '394 patent broadly describes “systems and methods for sample processing and data analysis,” which can include “nucleic acid sample processing and subsequent sequencing.” Ex. 1001, code (57). According to the Abstract, the resultant “sequence information may be analyzed with the aid of a computer processor, and the analyzed sequence information may be stored in an electronic storage location that may include a pool or collection of sequence information and analyzed sequence information generated from the nucleic acid sample.” *Id.*

⁶ Mr. West is a named inventor of the challenged patent. Ex. 1001, code (72). Mr. West co-founded Personalis and served as its Chief Executive Officer from August 2011 through December 2022. Ex. 2023 ¶ 3.

The '394 Specification more specifically discloses methods for “predicting, diagnosing, and/or prognosing a status or outcome of a disease or condition in a subject.” *Id.* at 55:38–50. In some embodiments, this involves the collection and analysis of sequence information derived from body fluid or tissue, including “from a subject suffering from a cancer.” *Id.* at 38:57–67, 55:51–57:8.

3. Challenged Claims

Petitioner challenges claims 1–19 of the '394 patent, of which only claim 1 is independent. Claim 1, recites:

1. A method for analyzing nucleic acid samples isolated from an individual, comprising:

- (a) generating a first subset of nucleic acid molecules from a first nucleic acid sample obtained from said individual;
- (b) conducting a first sequencing assay on said first subset of nucleic acid molecules to yield a first result comprising a first nucleic acid sequence wherein said first sequencing assay comprises whole genome sequencing;
- (c) generating a second subset of nucleic acid molecules from a second nucleic acid sample of said individual by contacting said second nucleic acid sample with a plurality of capture probes,
 - [i] wherein said plurality of capture probes hybridize to one or more polymorphisms which are based on or extracted from one or more databases of polymorphisms, observed in a population of one or more samples, or a combination thereof,
 - [ii] wherein said first sample or said second sample is obtained from said subject at a first time point;

- (d) conducting a second sequencing assay on said second subset of nucleic acid molecules to yield a second result comprising a second nucleic acid sequence;
- (e) using said plurality of capture probes to generate an additional subset of nucleic acid molecules from an additional nucleic acid sample isolated from an additional sample obtained from said subject at an additional time point different from said first time point;
- (f) conducting a third sequencing assay on said additional subset of nucleic acid molecules from said additional nucleic acid sample to yield a third result comprising a third nucleic acid sequence; and
- (g) generating a biomedical report that includes biomedical information of said subject, which biomedical information identifies the presence or absence of said one or more polymorphisms identified by said second result or said third result.

Ex. 1001, 78:54–79:24 (paragraphing added) (bracketed Roman numerals added as per POR, App’x A (claim listing)). Among the dependent claims, claim 12 (challenged only under Ground 3) specifies that the “capture probes are conjugated to beads,” claim 13 specifies that the “capture probes further comprise barcodes,” and claims 17 and 18 require that the “whole genome sequencing produces at least 750,000,000 [or 1,000,000,000] sequencing reads,” respectively. *See* POR, App’x A.

We further noted that, in the 01913 litigation, Patent Owner summarized claim 1 as

a method for analyzing nucleic acid samples isolated from an individual comprising whole genome sequencing nucleic acid molecules obtained from a first biological sample,
contacting a second set of nucleic acid molecules with a set of capture probes that hybridize to polymorphisms that are based

on or extracted from a database of polymorphisms or observed in a sample, or a combination thereof,
sequencing the second set of nucleic acid molecules,
repeating the capture probe hybridized sequencing on an additional sample from an additional time point, and
generating a biomedical report that identifies the presence or absence of the polymorphisms.

Ex. 1010 ¶ 30 (paragraphing added). Petitioner does not contest this characterization.

II. ANALYSIS

A. Legal Standards

“In an IPR, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

Petitioner contends that each claim of the ’394 patent is unpatentable as obvious under § 103(a). The Supreme Court in *KSR International Co. v. Teleflex Inc.*, 550 U.S. 398 (2007), reaffirmed the framework for determining obviousness set forth in *Graham v. John Deere Co.*, 383 U.S. 1 (1966). The *KSR* Court summarized the four factual inquiries set forth in *Graham* (383 U.S. at 17–18) that are applied in determining whether a claim is unpatentable as obvious under 35 U.S.C. § 103 as follows: (1) determining the scope and content of the prior art; (2) ascertaining the differences

between the prior art and the claims at issue; (3) resolving the level of ordinary skill in the art; and (4) considering objective evidence indicating obviousness or non-obviousness, if present. *KSR*, 550 U.S. at 406.

“[W]hen a patent ‘simply arranges old elements with each performing the same function it had been known to perform’ and yields no more than one would expect from such an arrangement, the combination is obvious.” *Id.* at 417 (quoting *Sakraida v. Ag Pro, Inc.*, 425 U.S. 273, 282 (1976)). But in analyzing the obviousness of a combination of prior art elements, it can also be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *Id.* at 418.

“[I]n considering the disclosure of a reference, it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). Moreover, a precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *KSR*, 550 U.S. at 418. Rather, “any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.” *Id.* at 420. Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (quotations and citations omitted).

B. Level of Ordinary Skill in the Art

In our Institution Decision, we provisionally adopted Petitioner’s definition of a person of ordinary skill in the art (“POSA”) as having:

- 1) an advanced degree (such as an M.D. or the equivalent of a Ph.D. in a life or physical sciences, engineering, or mathematics discipline) . . .
- 2) an understanding of various sequencing and next generation sequencing methods, sample preparation methods, targeted enrichment methods, bioinformatics methods for interpreting sequencing data, and methods for identifying genetic variants in a sample
- 3) an understanding of clinical applications of sequencing technology to conditions associated with nucleic acid variation, such as cancer, non-invasive prenatal testing, and transplant rejection . . . [and]
- 4) an understanding of cancer biology as it relates to mutations associated with tumor diagnosis, progression, and treatment, as well as awareness of the various types of mutations that can occur in and influence the course of a patient’s cancer.

DI, 15–19 (citing e.g., Pet. 8; Ex. 1020 ¶¶ 5, 37–91).

Patent Owner has not subsequently challenged the above definition. *See* Ex. 2031 ¶ 48 (Dr. Furneaux applying the above definition). For the reasons set forth in our Institution Decision, and because Petitioner’s proposed level is commensurate with the level of skill reflected in the prior art of record, we apply the same definition here. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978).

C. Claim Construction

“Only when a claim is properly understood can a determination be made . . . whether the prior art anticipates and/or renders obvious the claimed invention.” *Amazon.com, Inc. v. Barnesandnoble.com, Inc.*, 239

F.3d 1343, 1351 (Fed. Cir. 2001). In this proceeding, we interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2020). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.* Furthermore, we need only construe claim language that is in controversy, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999); *see also Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (applying *Vivid Techs.* in the context of an AIA trial proceeding). In this respect, Petitioner proposes constructions for the terms “subset” and “polymorphism,” argues that the “biomedical report” of claim step 1(g) is directed to printed matter not entitled to patentable weight, and raises the issue of whether the preamble is limiting. Pet. 19–23, 31. Patent Owner does not address the meaning of these terms and no express construction is required to resolve the controversy here.

We do address, below, the two claim phrases in dispute—“whole genome sequencing” and “capture probes.” As appropriate, we also discuss in this section, application of these terms to the asserted Leary reference.

1. “whole genome sequencing”

Claim step 1(b) recites “conducting a first sequencing assay on said first subset of nucleic acid molecules . . . wherein said first sequencing assay comprises whole genome sequencing.” Because of the substantial overlap in the parties’ arguments regarding the plain meaning of “whole genome

sequencing” and whether Leary discloses whole genome sequencing, we address both issues in this section.

Patent Owner and Dr. Furneaux contend that “whole genome sequencing,” or “WGS,” should be construed to mean, “**a non-targeted approach to sequencing all or almost all of a genome**”—a construction we preliminarily applied at institution. *See, e.g.*, POR 5–6; Sur-reply 9–12; DI 19–23; Ex. 2031 ¶¶ 50–52. Here, as articulated by Dr. Furneaux, Patent Owner contends that “whole” modifies “genome” and “thus indicates that the entirety—not a part or subset—of the genome must be sequenced.” Ex. 2031 ¶ 52.

Petitioner and Dr. Quackenbush, by contrast, contend that we should apply the plain meaning of whole genome sequencing, and present three, inter-related arguments as to their position as to what one skilled in the art would understand that meaning to be. *See generally*, Ex. 1225 ¶¶ 4–37. First, as is underscored by the parties’ briefing, Petitioner’s Ground 1 invites us to determine whether Leary’s PARE technique is a form of whole genome sequencing as understood by one of ordinary skill in the art at the time of the claimed invention. *See e.g.*, POR 5–9; Sur-reply 5–6.⁷

Petitioner contends that “[b]ecause the contemporaneous literature considered Leary’s technique to constitute WGS, no further construction is needed ‘to resolve the controversy’” Reply 5 (citing *Vivid Techs.* 200 F.3d at 803); Tr. 15:18–16:4. Considering the substantial briefing and oral argument regarding the term “whole genome sequencing,” however, we find it useful to undertake a formal construction. We, nevertheless, consider in this section

⁷ Petitioner also argues that to the extent Leary does not disclose whole genome sequencing, it would have been obvious to achieve whole genome sequencing using Leary’s PARE technique. *See* Reply 6–9; POR 10–12.

the parties' arguments regarding whether the contemporaneous literature understood Leary's technique as a form of whole genome sequencing.

Second, Petitioner contends, "to the extent express construction is required," we should interpret the plain meaning of whole genome sequencing as **"a non-targeted approach to generating sequencing reads and mapping those reads back to a reference genome (rather than a genome subset)."** Reply 5; Ex. 1225 ¶¶ 34–36. As a third alternative, Petitioner argues that to the extent we apply Patent Owner's preferred definition, the skilled artisan would understand the term to encompass a lower limit of 50% (or less) of the genome. *Id.* at 3–5.

We begin our analysis with the intrinsic record. At best, however, the Specification distinguishes whole genome sequencing from targeted sequencing, in which only a defined subset of genomic elements (e.g., exons) are sequenced. *See, e.g.*, Ex. 1001, 1:23–36, 5:57–59, 33:19–22, 75–76, Table 13 (distinguishing whole *genome* sequencing from whole *exome* sequencing).⁸ As such, the Specification is entirely consistent with Patent Owner's "a non-targeted approach to sequencing" and Petitioner's equivalent language, "a non-targeted approach to generating sequencing reads." Accordingly, there appears no dispute that whole genome sequencing does not target exons or other defined genomic elements.

With respect to the remainder of the parties' proposed definitions, Petitioner's construction recites, "mapping those reads back to a reference genome (rather than a subset)." The intended meaning of this phrase appears

⁸ The NIH National Human Genome Research Institute, defines an exome as "the sequence of all the exons in a genome, reflecting the protein-coding portion of a genome. In humans, the exome is about 1.5% of the genome." Ex. 3002, available at <https://www.genome.gov/genetics-glossary/Exome>.

to underscore the non-targeted, genome-wide nature of whole genome sequencing in not focusing on (or excluding) any subset of genetic elements or loci. Ex. 1225 ¶¶ 34–36; Tr. 14:21–15:14. While we have no disagreement with this interpretation, the evidence of record strongly suggests that whole genome sequencing requires something more than that a single sequencing read from a randomly selected genetic locus. Accordingly, and despite the agreement of Patent Owner’s counsel that the “specification defines no threshold for the amount of sequencing required,” we agree with Patent Owner’s contention that one of ordinary skill in the art would understand the term to require some “threshold amount of sequencing.” *See, e.g.*, Tr. 33:22–34:21.

With respect to Patent Owner’s position that whole genome sequencing requires “sequencing all or almost all of a genome,” the Specification nowhere defines whole genome sequencing in terms of the amount or percent of genome sequenced. At best it acknowledges that whole genome sequencing may include “low mean coverage regions,” and gaps described as “zero mean coverage regions.” *Id.* at 24:59–63, 28:33–42; *see also id.* at 7:18–19 (disclosing that the accuracy of the sequenced portion “may be greater than about 70%”). Absent further assistance from the Specification, we turn to the contemporaneous art and testimony of record to understand how those of skill in the art understood the term.

First, and consistent with the Specification’s recitation of “low” and “zero mean coverage regions,” we credit Dr. Quackenbush’s undisputed testimony that “sequencing of ‘all of the genome’” was unachievable at the time of the invention. Ex. 1225 ¶ 10 (citing Ex. 1222, 46:15–21, 37:17–24 (Dr. West’s testimony that it first became possible to sequence a human

genome without any gaps “*quite a long time after we filed these patents*”)); *see also id.* ¶ 10 n.1 (evidencing the first announcement of a gap free autosomal genome sequence in March 2022, completion of the Y chromosome sequence in August 2023); Tr. 32:22–33:4. Because sequencing “all” of a genome may have represented an aspirational goal for some applications, we decline to strike it from the definition. Moreover, because the Specification makes a clear, and repeated distinction between untargeted sequencing (e.g., whole genome sequencing) as compared to targeted sequencing approaches such as “whole exome sequencing,” the phrase “sequencing all or almost all of a genome” speaks more to the untargeted nature of the process rather than to the number of nucleotides sequenced. *See e.g.*, Ex. 1001, Ex. 1001, 1:23–36, 5:57–59, 33:19–22, 75–76 (Table 13). Consistent with this interpretation, the Specification’s references to “low mean coverage regions,” and “zero mean coverage regions” underscores that whole genome sequencing does not require that all or almost all of the nucleotide sequence of a genome must be determined, but that all or almost all of the genome must be available for sequencing (*i.e.*, in an untargeted manner). *See id.* at 24:59–63, 28:33–42.

a. The Range of Coverage Encompassed by Whole Genome Sequencing

The evidence of record demonstrates that one of ordinary skill in the art would have understood whole genome sequencing to encompass a wide range genome coverage, conveniently expressed as a percentage of the genome sequenced. *See* Ex. 2018, 155:15–21, 157:9–22, 161:21–162:1 (Dr. Quackenbush’s testimony regarding the scope of WGS in different contexts). On one hand, Patent Owner points to Ley as conducting whole

genome sequencing “to achieve the targeted level of 90% diploid coverage,” and Dewey⁹ as “providing 39.3x average coverage of 92% of known chromosomal positions.” Sur-reply 10–11 (citing Ex. 1012, 67; Ex. 2122, 2). Patent Owner further relies on the testimony of Dr. Furneaux and Mr. West that “90% sequence coverage was the accepted cutoff at the time of the invention.” *Id.* at 10 (citing Ex. 1222, 42:25–44:10; Ex. 1223, 97:23–98:5); *but see* Ex. 1223, 94:21–95:23 (Dr. Furneaux’s testimony that “all of this is a moving target”); Ex. 2031 ¶ 104 (Dr. Furneaux’s testimony that “‘whole genome sequencing’ has been used in the literature in different ways and in different contexts”).

Conversely, Petitioner and Dr. Quackenbush present credible evidence that in some applications whole genome sequencing includes coverage of only 5–10% of the genome. In particular, Petitioner points to Heitzer¹⁰ as teaching an application of whole genome sequencing encompassing only 10% of the genome. Ex. 1218, 2 (“whole-genome sequencing with a shallow sequencing depth of about 0.1x is sufficient for a robust and reliable analysis of copy number changes from single cells”), 7 (“[s]ingle-read whole genome plasma sequencing is performed with a shallow sequencing depth of approximately 0.1x”); Ex. 1225 ¶¶ 30–33. Likewise, the West patent,¹¹ assigned to Personalis, discloses the use of

⁹ Dewey, et al., Phased Whole-Genome Genetic Risk in a Family Quartet Using a Major Allele Reference Sequence, 7(9) PLoS Genetics, September 2011. (Ex. 2122).

¹⁰ Heitzer, *et al.*, “Tumor-associated copy number changes in the circulation of patients with prostate cancer identified through whole-genome sequencing,” 5:30 Genome Med. 1–16 (2013). Ex. 1218.

¹¹ West, U.S. Patent No. 11,584,968, issued Feb. 21, 2023. Ex. 1212.

“untargeted sequencing data (e.g., a low coverage whole genome sequencing data),” and cites to Pasaniuc,¹² which describes “sequencing libraries appropriate for whole-genome sequencing” that are sequenced “at ultra-low coverage (0.1-0.5x[*i*])” of 5–10%.¹³ Ex 1212, code (56), 26:6–9; Ex. 1213 (emphasis omitted); Ex. 1225 ¶ 23.

Although the teachings of Heitzer, the West patent, and Pasaniuc plausibly suggest we interpret “sequencing all or almost all of the genome” not as a measure of minimum coverage, but as reiterating the non-targeted aspect of whole genome sequencing in not selecting for or against any defined genetic elements, we focus here on whether whole genome sequencing requires some minimum level of genomic coverage. Relevant to our analysis, the parties calculate that the asserted Leary reference sequenced about 60% or 53% of the genome using the PARE approach.¹⁴ See Reply 6; Sur-reply 14; Ex. 1225 ¶ 66; Tr. 14:14–17, 16:5–10, 36:16–25. Accordingly, whereas the application of whole genome sequencing at the shallow depth, low, or ultra-low coverage disclosed in Heitzer, the West patent, and Pasaniuc, might apply in certain contexts, the controversy before

¹² Pasaniuc, *et al.*, “Extremely low-coverage sequencing and imputation increases for genome-wide association studies,” 44(6) Nat. Genet. 631–635 (2012). Ex. 1213.

¹³ Our reliance on Pasaniuc is tempered by its disclosure that it conducted “a whole-*exome* study,” as “an empirical demonstration.” *Id.* at Abstract (emphasis added).

¹⁴ The difference in percentage appears to reflect an evolving understanding of the size of the reference genome, which forms the denominator for the calculations. See, e.g., Reply, 2–3; Tr. 14:14–17; Ex. 1225 ¶¶ 12–20 (Dr. Quackenbush’s testimony that one of ordinary skill in art at the relevant time understood the reference genome to be about 2.86 million bases, rather than the 3.2 billion used in Dr. Furneaux’s calculations).

us does not require a precise determination of the minimum genome coverage required for whole genome sequencing.¹⁵ Rather, for the purpose of addressing the controversy as posed by Patent Owner, it is sufficient that we consider the more general teachings of the Harris¹⁶ and Chandratillake¹⁷ patents regarding the range of coverage encompassed by whole genome sequencing.

The Harris patent issued from a series of continuation applications first filed on August 29, 2014, and claims benefit of priority to a U.S. Provisional application filed on August 30, 2013. Ex. 1211, code (63). The Harris patent identifies Personalis as both Applicant and Assignee and lists as an inventor John West, who is also named as an inventor of the '394 patent. Ex. 1211, codes (71–73); Ex. 1001, code 72. The Chandratillake patent issued from a series of continuation applications first filed on October 3, 2014, and claims benefit of priority to a U.S. Provisional application filed on October 3, 2013. Ex. 1214, codes (71, 73, 63). Like the Harris patent, it also identifies Personalis as Applicant and Assignee, and shares named inventors (Gemma L. Chandratillake, Sarah K. Garcia, and Richard Chen), with the '394 patent at issue here. Ex. 1214, codes (71–73); Ex. 1001, code 72.

By comparison, the '394 patent was first filed on December 27, 2013, and claims benefit of priority to a provisional application filed on January

¹⁵ Heitzer, the West patent, and Pasanuic, however, render irrelevant Mr. Furneaux's citation to McBride as evidence that Leary disclosed "only low coverage sequencing of the cancer genome." *See* Ex. 2031 ¶ 108.

¹⁶ Harris et al., US 11,456,058 B2, issued Sept. 27, 2022. Ex. 1211.

¹⁷ Chandratillake et al., US 11,640,405 B2, issued May 2, 2023. Ex. 1214.

17, 2013. Ex. 1001, code (60). Whether the Harris and Chandratillake patents qualify as prior art to the '394 patent, they are at least closely contemporaneous with it. Moreover, to the extent one of ordinary skill in the art would have placed a lower limit on the amount of coverage required by whole genome sequencing, we see no reason why that limit would be reduced in the year or so following the critical date, particularly given the improvements in cost and availability of large-scale sequencing over time. *See generally*, Pet. 10–16. In sum, we agree with Dr. Quackenbush that the Harris and Chandratillake patents reasonably encompass what one of ordinary skill would have understood by the term whole genome sequencing. *See* Ex. 1225 ¶¶ 21–22, 24.

The Harris and Chandratillake patents state:

Untargeted sequencing (i.e., whole genome sequencing) can determine the complete DNA sequence of the genome at one time. Untargeted sequencing (i.e., whole genome sequencing or the non-exonic portion of whole exome sequencing) can cover sequences of almost about 100 percent, or about 95%, of the sample's genome. **In some cases, the untargeted sequencing (i.e., whole genome sequencing or non-exonic portion of the whole exome sequencing) can cover sequences of the whole genome of the nucleic acid sample of about or at least about 99.999%, 99.5%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, 85%, 84%, 83%, 82%, 81%, 80%, 79%, 78%, 77%, 76%, 75%, 74%, 73%, 72%, 71%, 70%, 69%, 68%, 67%, 66%, 65%, 64%, 63%, 62%, 61%, 60%, 59%, 58%, 57%, 56%, 55%, 54%, 53%, 52%, 51%, or 50%.**

Ex. 1211, 17:33–47 (emphasis added); Ex. 1214, 29–43 (same). The Harris patent further explains that “the output [of WGS] can have a coverage of about or at least about . . . 50% of the whole genome of the nucleic acid sample from a subject.” Ex. 1211, 17:48–55.

We also find the Harris patent particularly relevant to the present controversy regarding Leary in reciting that in some embodiments,

the whole genome sequencing data comprises mate-pair reads. In another embodiment, the paired-end reads have insert-sizes of larger than about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0, 12.0, 13.0, 14.0, 15.0, 16.0, 17.0, 18.0, 19.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0, 90.0, or 100.0 kilobasepairs. In another embodiment, the ***mate-pair reads have insert-sizes of*** larger than about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, ***1.4***, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0, 12.0, 13.0, 14.0, 15.0, 16.0, 17.0, 18.0, 19.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0, 90.0, or 100.0 ***kilobasepairs***.

Id., 3:37–49 (emphasis added). As explained by Dr. Quackenbush, as in Harris, “Leary’s whole-genome sequencing comprised mate-pair reads with insert lengths of 1,400 bp (1.4 kilobasepairs).” Ex. 1225 ¶ 22 n.2 (cited at Reply 4).

We are not persuaded by the arguments of Patent Owner’s counsel that we should dismiss the teachings of the Harris and Chandratillake patents. *See generally*, Tr. 30:24–31:22, 33:5–35:4, 37:1–38:2. Consistent with the inventor’s own words in the Harris and Chandratillake patents, the record as a whole supports Petitioner’s view that one of ordinary skill in would have understood the 60% or 53% genomic coverage achieved with Leary’s mate-paired tag/PARE analysis to be sufficient for whole genome sequencing.

In sum, we construe whole genome sequencing as a non-targeted approach to sequencing all or almost all of a genome, which encompasses determining the nucleotide sequence covering about 50% or more of a genome.

b. Whether Leary Discloses Whole Genome Sequencing

i. Overview of Leary (Exhibit 1002)

According to Leary, “[v]irtually all tumors of clinical consequence are thought to have rearranged DNA sequences resulting from translocations and copy number alterations, and these sequences are not present in normal human plasma or nontumor tissues.” Ex. 1002, 5. Leary describes methods of analyzing patient-specific translocations in solid tumors using the PARE (personalized analysis of rearranged ends) approach, which involved comparing information derived from tumor samples with matched normal samples. *Id.* at Abstract, 1. In obtaining this information,

[g]enomic DNA from each [of four tumor samples] was purified, sheared, and used to generate libraries with mate-paired tags ~1.4 kb apart. Libraries were digitally amplified by emulsion polymerase chain reaction (PCR) on magnetic beads (21), and 25-bp mate-paired tags were sequenced using the sequencing-by-ligation approach (15, 22). An average of 198.1 million 25-bp reads was obtained for each sample, where each read aligned perfectly and was uniquely localized in the reference human genome (hg18), resulting in 4.95 Gb of mappable sequence per sample. An average of 40 million mate-paired reads, where both tags were perfectly mapped to the reference human genome, was obtained for each sample. The total amount of genome base pairs covered by the mate-paired analysis (that is, distance between mate-paired tags × number of mate-paired tags) was 53.6 Gb per sample, or an 18-fold physical coverage of the human genome.

Id. at 1.

Using the mate-paired tag/PARE approach, Leary “identified 57 regions containing putative somatic rearrangements.” *Id.* at 3. “For confirmation, [Leary] designed primers to 42 of the paired-end regions and used them for PCR spanning the putative breakpoints. Thirty-five of these (83%) yielded PCR products of the expected size in the tumor samples but

not in the normal samples (Fig. 2 and table S3).” *Id.* Leary reports that “Sanger sequencing of seven PCR products confirmed the rearrangements in all cases tested. Although there was variation in the number of detected alterations per sample (range, 7 to 21), all four tumor samples were found to have at least four bona fide somatic rearrangements through this approach.” *Id.*

Leary also demonstrates that rearranged tumor markers identified using PARE could be detected and quantified in clinical samples, *e.g.*, in serial patient plasma samples taken before and after tumor resection and chemotherapy. *Id.* at 4. Leary reports, for example, that

[d]igital PCR analyses of plasma samples from patient Hx402 indicated that rearranged tumor DNA was detectable at all six time points analyzed throughout the course of treatment (Fig. 4B). The fraction of mutant DNA present in plasma samples was 37% just before surgery but decreased to 14% 1 day after resection of the primary tumor. The mutant DNA fraction decreased further after chemotherapy and subsequent removal of metastatic lesions from the right lobe of the liver.

Id. In Figure 4B (referenced above), Leary graphically displays the fraction of rearranged DNA in a patient’s plasma over time, and in light of the patient’s chemotherapy and surgical history. *Id.* at 4, Fig. 4. Referencing Figure 4B, Leary states that “[d]igital PCR analyses of plasma samples from patient Hx402 indicated that rearranged tumor DNA was detectable at all six time points analyzed throughout the course of treatment.” *Id.* Leary concludes that, “[g]iven current enthusiasm for the personalized management of cancer patients, PARE affords a timely method for uniquely sensitive and specific tumor monitoring.” *Id.* at 5.

ii. Analysis

According to Patent Owner, Chan refers to Leary's work as having a "targeted nature" inconsistent with a definition of whole genome sequencing as "a non-targeted approach to sequencing all or almost all of a genome." Sur-reply 9, 12–13 (citing Ex. 1002, 1, 5; Ex. 2031 ¶¶ 93–96; Ex. 1020 ¶ 227 (Dr. Quackenbush quoting Patent Owner's relied-upon passage from Chan)). In a similar respect, Dr. Furneaux describes Leary as disclosing "a breakpoint *mapping* technique that only uses the sequences of the mate-pair ends to identify the putative *location* of [a] breakpoint rather than determining the order of the nucleotides in all (or almost all) of the genome." Ex. 2031 ¶ 108 (citing Ex. 1015, 9); *see* Ex. POR 9. We do not find Patent Owner's arguments availing for the reasons set forth at pages 39–40 of our Institution Decision and as discussed in paragraphs 58–63 of Dr. Quackenbush's second Declaration (Ex. 1225), which we adopt.

In short, that Leary analyzed genomic sequencing data for the purpose of identifying and/or mapping "rearranged DNA sequences resulting from translocations and copy number alterations," has no bearing on whether the underlying generation of genomic sequence information using a PARE approach comprised WGS. *See* Ex. 1002, 5; *see also* Ex. 1030, 10 (describing Leary as disclosing "identification of rearrangements using whole-genome sequencing"). Rather, because Leary's sequencing library was generated from randomly sheared genomic DNA, the resulting genomic sequences reflected a non-targeted approach. *See e.g.*, DI, 22 & n.10 (explaining that "Leary's PARE studies began with sheared genomic DNA, and presumably resulted in non-targeted, randomly-distributed sequencing reads throughout the genome"). Read in context, Chan refers to Leary's work as having a "targeted nature" that "can provide only a partial glimpse

of the tumor genome in the plasma of cancer patients.” Ex. 1008, 212. As such, we agree with Dr. Quackenbush that

this passage of Chan is comparing its whole genome sequencing of plasma to Leary’s follow-up analysis, which involved PCR-based capture followed by targeted resequencing. Chan’s reference to “targeted” sequencing in Leary is not a reference to Leary’s initial polymorphism discovery using its PARE whole genome sequencing assay, which was performed on tissue DNA, not the plasma.

Ex. 1225 ¶ 60. Chan’s reference to “targeted nature,” thus, refers to Leary’s overall analysis, and not the initial PARE approach to generating genomic sequence data. *See* Ex. 1008, 211–212.

c. Whether the Contemporaneous Literature Understood Leary to Disclose Whole Genome Sequencing

We also address Petitioner’s contention that the contemporaneous literature considered Leary’s technique to constitute whole genome sequencing. *See* Reply 5; Resp. 2–3; Ex. 1225 ¶¶ 22–29, 68. As noted in our Institution Decision, Forshew¹⁸ expressly describes Leary (reference 18 in Forshew) as disclosing “identification of rearrangements using whole-genome sequencing.” DI 21–22 (quoting Ex. 1030, 11). Petitioner further points to Diehn¹⁹ as expressly disclosing that Leary teaches whole genome sequencing. Reply 4; Ex. 1225 ¶ 26. Diehn is a U.S. Patent Application Publication indicating the benefit of a provisional application filed on March 15, 2013. Ex. 1112, code (60). As with Harris and Chandratillake, Diehn is

¹⁸ Forshew, et al., “Noninvasive Identification and Monitoring of Cancer Mutations by Targeted Deep Sequencing of Plasma DNA,” 4 *Science Translational Medicine* 136ra68 (2012). Ex. 1030.

¹⁹ Diehn et al., US 2022/0195530 A1, publ. June 23, 2022. Ex. 1112.

reasonably contemporaneous with the '394 patent. In a background section discussing sequencing techniques for analyzing cancer-derived cell-free DNA (cfDNA), Diehn includes the following with express citation to Leary: “Other studies have proposed identifying patient-specific chromosomal rearrangements in tumors via whole genome sequencing (WGS), followed by breakpoint qPCR from cfDNA (Leary et al. (2010) Sci. Transl. Med. 2: 20ra14.” *Id.* ¶ 3. We find Diehn’s statement pertinent to understanding how one of ordinary skill art understood the teachings of Leary.

In response, Patent Owner points to Feldman²⁰ as evidence that whole genome sequencing excludes Leary’s PARE approach. POR 9. As reproduced by Dr. Furneaux, Feldman states:

Whole-genome sequencing currently demands resources beyond the reach of most cancer investigators. Transcriptome sequencing has been used to identify novel fusion genes derived from chromosomal translocations in solid tumors. In lymphomas, however, translocations often do not produce a fusion gene and, in some cases, disrupt a putative tumor suppressor gene. Transcriptome sequencing is unlikely to detect such translocations. ***Therefore, we used a mate-pair genomic DNA sequencing approach for lymphoma translocation discovery.***

Ex. 2031 ¶ 105 (quoting Ex. 2008, 918). Feldman, however, continues:

In this resource-efficient approach, only the ends of approximately 5-kb DNA fragments are sequenced, such that we obtained approximately 7 times the bridged coverage of the genome despite base coverage of only approximately 0.35 times, requiring only one lane of an Illumina flow cell. Our binary indexing algorithm maximizes computer memory and

²⁰ Feldman et al., “Discovery of recurrent t(6;7)(p25.3;q32.3) translocations in ALK-negative anaplastic large cell lymphomas by massively parallel genomic sequencing,” 117(3) *Blood*, January 20, 2011. Ex. 2008.

speed while allowing the reference genome and its reverse complement to be stored in RAM. This allows mapping approximately 2×10^8 sequences to a 3×10^9 -nucleotide genome in one day. ***This approach is much faster and less labor-intensive than approaches such as long-distance inverse PCR. In addition, the whole-genome coverage and 5-kb resolution are more informative than karyotyping or FISH.***

Ex. 2008, 918. Feldman’s apparent teaching that “a mate-pair genomic DNA sequencing approach,” (i.e., the general approach taught by Leary), provides “[w]hole-genome coverage,” is somewhat at odds with Dr. Furneaux’s contention that Feldman “specifically *exclude[s]* Leary’s PARE approach when describing whole genome sequencing.” Ex. 2031 ¶ 105. We find more persuasive that Forsheew and Diehn expressly refer to Leary as disclosing whole genome sequencing.

Patent Owner further points to Meyerson’s²¹ Table 1 (“Whole-genome sequencing studies of cancer”), and a passage from Newman²² (a 2014 article regarding the quantitation of circulating tumor DNA (ctDNA) in plasma co-authored by two of Foresight’s founders), as evidence that these references do not expressly cite to Leary when discussing whole genome sequencing. *See* POR 9 (citing Ex. 1031, 687, Table 1; Ex. 2152, 7; Ex. 2031 ¶¶ 106–7). We are not persuaded that the absence of a citation to Leary in these contexts indicates that one of ordinary skill did not consider Leary to disclose whole genome sequencing. To the contrary, Dr. Quackenbush

²¹ Meyerson, et. al., “Advances in understanding cancer genomes through second-generation sequencing,” 11 *Nature Reviews* 685–696 (2010). Ex. 1031.

²² Newman, et al., “An ultrasensitive method for quantitating circulating tumor DNA with broad patient coverage,” 20(5) *Nat Med.* 548–554 (2014). Ex. 2152.

testifies persuasively that “Meyerson’s list does not purport to be exhaustive,” and, in fact, “cites Campbell 2008 and Stephens 2009, each shallow paired end whole-genome sequencing studies” similar to Leary. *See* Ex. 1225 ¶¶ 28–29. We further credit Dr. Quackenbush’s explanation that it is unsurprising that the paragraph in Newman on which Patent Owner relies does not include Leary, because it “is plainly discussing studies focused on measurements on ctDNA,” whereas “[t]he whole genome sequencing in Leary is performed on DNA derived from tissue.” *Id.* ¶ 30–32; Sur-reply 2–3. Finally, we do not find persuasive Patent Owner’s conjecture that because Leary’s authors used the term whole genome sequencing in 2012 (Ex. 2183) but not in Leary 2010 (Ex. 1002), the asserted 2010 reference must not have used whole genome sequencing. Sur-reply 9; *see* Ex. 2183, Abstract (describing “Whole-genome analyses of DNA from the plasma”). Instead, we credit Dr. Quackenbush’s testimony that in the later paper, the authors are saying:

having failed to detect a rearrangement by simply doing digital PCR, let’s go back and do PARE, right. Let’s start by doing whole genome sequencing, like we did in PARE where we found de novo translocations, and look to see what happened, right.

So they’re not replacing PARE with whole genome sequencing. They’re replacing digital PCR with PARE, which is whole genome sequencing. So I think you’ve got the interpretation of this kind of backward,

Ex. 2188, 215:8–216:10; *see also id.* at 217:10–20 (Dr. Quackenbush’s testimony that one of ordinary skill in the art would understand Leary to disclose whole genome sequencing).

d. Conclusions Regarding the Meaning of Whole Genome Sequencing and Its Application to Leary

In sum, and considering all the evidence of record, we construe whole genome sequencing as a non-targeted approach to sequencing all or almost all of a genome, which encompasses determining the nucleotide sequence covering about 50% or more of a genome. We further find that the application of the mate-paired tag/PARE technique in Leary satisfies this definition as it is a non-targeted approach, which, the parties agree, resulted in the sequencing of about 53% or 60% of the genome (depending on the calculation applied). *See* Reply 6; Sur-reply 14; Ex. 1225 ¶ 66. Finally, and irrespective of the term’s precise bounds, we find, on balance, that the art of record evidences that the POSA would have recognized that Leary discloses whole genome sequencing.

2. “capture probes”

Claim step 1(c) involves “generating a second subset of nucleic acid molecules . . . by contacting said second nucleic acid sample with a plurality of capture probes, wherein said plurality of capture probes hybridize to one or more polymorphisms.” The construction of “capture probes” in this phrase is particularly relevant as we must address whether the term, properly construed, encompasses Leary’s PCR primers. POR 27–31 (Patent Owner’s argument that Leary does not teach or suggest “capture probes”). Indeed, Patent Owner contends “Foresight’s entire challenge to the ’394 patent hinges on a POSA conflating Leary’s PCR primers with the claimed ‘capture probes.’” POR 27; *see also* Tr. 19:19 (“[a] central dispute is the construction of capture probes”).

Patent Owner argues that one of ordinary skill in the art would have understood “capture probes” to mean “a nucleic acid sequence that

hybridizes to target DNA, either in solution or on a solid support, so that one can physically capture and isolate the target DNA”—a definition, it contends, excludes PCR primers, such as those employed by Leary. *See generally*, POR 16–31; Sur-reply 2–9.

For the purpose of institution, we found it sufficient to provisionally adopt Petitioner’s proposed definition of “capture probes” as encompassing at least, “PCR primer capture probes and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits).” *See* DI 23–29. For the reasons set forth in our Institution Decision, and as further addressed below, we apply that definition here—albeit with a minor modification in wording. In particular, we credit Patent Owner’s argument that the phrase “PCR primer capture probe” is not a recognized term of art and its use in our construction “would improperly inject ambiguity into the claim.” POR 18–19 (citing, e.g., Ex. 2018, 166:10–169:1, 171:16–175:2; *Liquid Dynamics Corp. v. Vaughan Co.*, 355 F.3d 1361, 1367 (Fed. Cir. 2004)). Although we recognize Petitioner’s use of “PCR primer capture probe” as a shorthand for its position that, as read in light of the Specification, PCR primers are a species of, and/or indistinguishable from, the genus of capture probes, Patent Owner’s point is well taken. Accordingly, we clarify the meaning of “capture probes,” as used in the instant Specification, to encompass PCR primers and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits). We do not view this modification as altering the substance of our provisional construction or the analysis in our Institution Decision.

In considering how one of ordinary skill in the art would have understood “capture probes” in the context of the ’394 patent, we look first

to the Specification—“the single best guide to the meaning of a disputed term.” *In re Abbott Diabetes Care Inc.*, 696 F.3d 1142, 1149 (Fed. Cir. 2012) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1315 (Fed. Cir. 2005) (en banc)). The ’394 patent’s Specification provides no express definition of “capture probes,” but does, as Patent Owner points out, “describe[] several features of capture probes.” *See* POR, 16–17; *see also*, Tr. 20:1–21:19, 28:3–9 (“our specification does not provide a definition. It provides some description, but not a complete description”).

With respect to structure, the ’394 patent teaches that, “capture probes may comprise *one or more nucleotides . . . or 1000 or more nucleotides.*” Ex. 1001, 31:35–43 (emphasis added); *id.* at 41:33–34 (“The capture probes . . . may comprise one or more nucleotides.” In other words, capture probes within the scope of the ’394 patent encompasses a nucleic acid range from a single nucleic acid to one of arguably infinite length.²³

The Specification further discloses that a capture probe may *optionally* comprise one or more linkers and/or one or more labels. Ex. 1001, 31:4–12 (“The capture probe may further comprise one or more linkers. The capture probes may further comprise one or more labels. The one or more linkers may attach the one or more labels to the nucleic acid binding site.”). According to the Specification, the optional linker and/or label component of a capture probe encompasses a wide range of suitable materials:

Examples of linkers include, but are not limited to, hydrazone, disulfide, thioether, and peptide linkers. The linker may be a peptide linker. The peptide linker may comprise a proline

²³ Patent Owner agrees that the Specification encompasses capture probes having a length of one nucleotide. POR 23 (citing Ex. 2031 ¶ 69; Ex. 2027 (Gellert et al., “Helix Formation by Guanylic Acid,” 48 Proc. Nat. Acad. Sci. 2013–2018 (1962))).

residue. The peptide linker may comprise an arginine, phenylal[a]nine, threonine, glutamine, glutamate, or any combination thereof. The linker may be a heterobifunctional crosslinker.

Id. at 43:5–11; *see also id.* at 42:49–51 (“Suitable linkers comprise any chemical or biological compound capable of attaching to a label, primer, and/or capture probe disclosed herein.”)

With respect to “label” (interchangeable with “detectable label”), the Specification further teaches that these term’s “generally refer() to any chemical moiety attached to a nucleotide.” *Id.* at 14:22–29. More particularly,

labels include, but are not limited to, detectable labels, such as radioisotopes, fluorophores, chemiluminophores, chromophore, lumiphore, enzymes, colloidal particles, and fluorescent microparticles, quantum dots, as well as antigens, antibodies, haptens, avidin/streptavidin, biotin, haptens, enzymes cofactors/substrates, one or more members of a quenching system, a chromogens, haptens, a magnetic particles, materials exhibiting nonlinear optics, semiconductor nanocrystals, metal nanoparticles, enzymes, aptamers, and one or more members of a binding pair.

Id. at 23:23–36. In addition to the above, the Specification further discloses that at least some linker molecules may also be labels. *See also id.* at 14:29–42 (further exemplifying labels as including “linker molecules” such as “biotin, avidin, streptavidin, HRP, protein A, protein G, antibodies or fragments thereof”). Illustrating the breadth of this term, the Specification “envision[s] that a change in mass may be considered a detectable label.” *Id.* at 14:42–47.

The Specification also refers to bead-bound nucleic acid molecules “associated with a nucleic acid molecule.” *Id.* at 14:18–19; 56–58. Relatedly, the Specification teaches that “one or more subsets of nucleic acid

molecules may be produced by contacting a nucleic acid sample with one or more beads, capture probes, labels, or a combination thereof.” *Id.* at 32:12–15. “The capture probes, primers, labels, and/or beads *may comprise one or more nucleotides*. The one or more nucleotides may comprise RNA, DNA, a mix of DNA and RNA residues or their modified analogs such as 2'-OMe, or 2'-fluoro (2'-F), locked nucleic acid (LNA), or abasic sites.” *Id.* at 41:32–37 (emphasis added). As such, the Specification instructs that the structure of capture probes (as well as primers, labels, and/or beads) encompasses a nucleic acid of any length, with or without the addition of other adducts—which, themselves, may comprise a single nucleotide and/or a vast array of other entities.

In addition to the above, the Specification teaches that a capture probe may also possess the functional attributes of binding or hybridization. Consistent with the plain language of step 1(c)(i), “wherein said plurality of capture probes hybridize to one or more polymorphisms,” the Specification teaches that “[t]ypically, the capture probe comprises a nucleic acid binding site” and “may comprise a nucleic acid binding site that hybridizes to at least a portion of the one or more nucleic acid molecules or variant or derivative thereof in the sample or subset of nucleic acid molecules.” *Id.* at 31:7–8, 24–27. According to the Specification, “a capture probe hybridized nucleic acid refers [to] a capture probe associated with a nucleic acid molecule. The capture probe and the nucleic acid are in contact with each other.” *Id.* at 7:57–64, 14:48–59; *see also id.* at 22:65–23:9 (discussing genomic features/regions that may be hybridized by capture probes).

The Specification discloses that, “[c]apture probes may hybridize to *one or more* genomic regions” (*id.* at 22:65–23:1 (emphasis added)), but

does not expressly define the size of a nucleic acid binding site, nor the size of a genomic region that needs to hybridize with the capture probe, nor binding conditions, nor any particular binding affinity for the capture probe.²⁴ All that is required functionally, is that the capture probe “hybridizes to at least a portion of the one or more nucleic acid molecules or variant or derivative thereof in the sample or subset of nucleic acid molecules.” *Id.* at 31:25–27.

In short, all the Specification requires of capture probe function is that it hybridize to one or more nucleic acids. And as articulated at the institution stage, “[u]nder some (unspecified) conditions, and at some (unspecified) binding affinity, a single nucleic acid would be expected to hybridize with its complement and thereby meet the “capture probe” limitation as set out in the ’394 patent Specification.” DI, 44 (Jenks J., concurring).

In provisionally adopting Petitioner’s proposed definition in our Institution Decision, we first addressed Patent Owner’s contention that the ’394 patent “refers to capture probes and primers as separate and distinguishable from one another.” *See* DI 25 (citing e.g., Ex. 1001, 23:23–36; 32:56–60, 41:33–36). Considering the cited passages and, moreover, the entirety of the Specification, we were unable to discern any teaching as to *how* capture probes and primers might be “separate and distinguishable from

²⁴ Given that the Specification teaches that a capture probe may be as short as one nucleotide, we might infer that the minimum size of its binding site and hybridized genomic region is also one nucleotide. *See* Ex. 1001, 31:35–43, 41:33–34; *see also* POR 23 (Patent Owner’s argument that “[t]he specification correctly includes capture probes having a length of one nucleotide because there are instances where a single nucleotide can function as a capture probe”).

one another,” as Patent Owner contends. *Id.* at 25–26. We noted, in particular, that

the Specification evinces no physical characteristics differentiating capture probes from primers. Both comprise from as few as “one or more nucleotides” to “1000 or more nucleotides” (Ex. 1001, 31:35–53, 40:28–54); both include nucleotides “compris[ing] RNA, DNA, a mix of DNA and RNA . . . modified analogs . . . locked nucleic acid (LNA), or abasic sites” (41:32–36); both may—or may not—comprise a detectable label (*id.* at 14:22–47, 23:23–36, 31:54–32:5, 40:29–35, 41:38–42:3); and both may—or may not—comprise a linker (*id.* at 31:8–9, 31:54–67, 40:29–31, 41:15–26, 42:49–43:11), wherein said linker can also be a detectable label (*id.* at 14:36–42).

DI 26.

With respect to function, we reasoned that because the Specification provides no physical difference between capture probes and primers “it would appear that the same polynucleotide could be employed as an amplification primer and a capture probe.” *Id.* at 27 (citing Prelim. Resp. 23). We likewise agreed with Petitioner that “none of the cited passages [by Patent Owner] describe ‘primers’ and ‘capture probes’ as mutually exclusive or precludes a similarly structured nucleic acid from performing each function.” *Id.* at 23; *see also* Reply 10–11 (citing POR 19, Ex. 1001, 23:23–26; Ex. 1225 ¶ 40).

We also noted Dr. Quackenbush’s testimony that dependent claim 13’s requirement that “capture probes further comprise barcodes,” typically indicated the use of PCR primers, but not hybrid capture baits. DI 23–24 (citing Ex. 1020 ¶¶ 104–105). And although Patent Owner responds that “barcodes can also be used on capture probes” (POR 24 (citing Ex. 2031 ¶ 70; Ex. 2028, 251)), this is not quite on point as Dr. Quackenbush and

Mr. West subsequently confirmed that, while the use of barcoded PCR primers are well known, where bar codes are used in the context of biotinylated oligonucleotide bait systems, *the barcodes are not part of the capture probes*, rather, the barcodes are “part of the [captured DNA] molecule that’s being sequenced.” Ex. 1222, 167:20–170:4; Ex. 1225 ¶ 41.

Patent Owner contends that its proffered definition of “capture probes” as meaning “*a nucleic acid sequence that hybridizes to target DNA, either in solution or on a solid support, so that one can physically capture and isolate the target DNA,*” excludes PCR primers, such as those employed by Leary. *See generally*, POR 16–31; Sur-reply 2–9. In support of its construction, Patent Owner points first to the intrinsic record. Patent Owner argues that the Specification “treats capture probes and PCR primers as different types of nucleic acids,” that it separately describes the two terms, and, for some embodiments, “describe[s] the use of capture probes as a separate and distinct step from using PCR primers.” POR 19 (citing Ex. 1001, 31:4–32:5, 40:29–41:36, 23:23–26, 32:56–60, 64:29–43, 65:38–50, 66:53–67). Patent Owner made much the same argument prior to Institution, but again, does not adequately describe any physical difference between the capture probes and PCR primers in its citations, nor persuade us that these entities, so defined in the Specification, are not functionally interchangeable. *See* DI 25–27 (concluding that “reading the Specification, it would appear that the same polynucleotide could be employed as an amplification primer and a capture probe in Patent Owner’s referenced embodiment(s)”).

Patent Owner further argues that during the prosecution history of the '394 and '033 patents, the Examiner²⁵ agreed “that ‘pulldown probes’ are not PCR primers” and, in conducting prior art searches, the Examiner used the terms “capture” and “pull-down” synonymously, without including “primers” or “PCR” as search terms. POR 20 (citing Ex. 2004, 14, 17, 25; Ex. 2005, 19, 24, 31; Ex. 2024, 3–8, Ex. 2025, 3–25). But as previously noted, when the Examiner agreed that “pulldown probes” are not PCR primers, “the cited applications involved the claim limitation ‘pulldown probe,’ and claims reciting ‘capture probe’ were not before the Office.” DI 27–28 (“Although the cited interchange does not involve the claim term “capture probe,” it shows the Examiner understood that ‘probes’ encompassed ‘amplification primers’”). As such, Patent Owner’s argument remains of limited value.

As to Patent Owner’s assertion that the Examiner did not include “PCR” or “primers” in conducting prior art searches—but did apply “capture” and “pull-down” as synonyms—we note that the Examiner’s arguably synonymous search terms further encompassed “enrichment.” *See* Ex. 2024, 3–8 (“(enrich\$5 captur\$5 pull-down pulldown)”); Ex. 2025, 3–25 (same). As Patent Owner admits, the prior art recognized PCR as an enrichment technique. *See, e.g.*, POR 22 (citing Ex. 1006, 375), 24 (citing Ex. 1006, 378–80), 26 (“a POSA would understand that capture probes and PCR primers are different enrichment techniques”). As such, we might infer that the Examiner understood the search parameters to encompass PCR and

²⁵ Primary Examiner Kaijiang Zhang examined the '394 patent, the '337 application, and the '215 application. *See* Ex. 1001, code (74); Ex. 1078, 1; Ex. 2004, 2; Ex. 2005, 12.

PCR primers. Moreover, even if the Examiner had focused solely on the smaller genus of pull-down probes, this would still read on a claim more broadly reciting “capture probes,” properly construed. In any event, Patent Owner’s argument does not overcome the clear evidence in the Specification that the inventors chose to so broadly describe capture probes and PCR primers as to render the terms interchangeable. At a minimum, we find no persuasive evidence in the Specification or cited prosecution history that the scope of the term “capture probes” necessarily excludes PCR primers.

We also find persuasive Petitioner’s evidence that one of ordinary skill in the art would not have understood “capture probes” as synonymous with “pulldown probes.” Reply 11. In this respect, Mr. West and Dr. Quackenbush agree that a “pulldown probe” refers to a biotinylated oligonucleotide bait used in a “pulldown assay,” wherein the interaction between biotin and streptavidin attached to, e.g., a magnetic bead, “pulls down” or precipitates a molecule of interest. *See* Ex. 1222, 83:18–85:22; Ex. 1225 ¶ 39, 48. Patent Owner’s expert, however, testified that “capture probes” was a broader term, encompassing at least array-based hybrid capture, solution-based hybrid capture, guanine nucleotide—as taught by Gellert (Ex. 2027), Southern blot probes, and the zip-code PCR/ligase micro detection array probes taught by Gerry.²⁶ *See* Ex. 2031 ¶¶ 65–70. Moreover, focusing on the intrinsic evidence, Petitioner notes:

The specification is explicit that [biotin and streptavidin] are considered “linkers” or “labels,” Ex1001, 14:22-46, 23:23-36, 31:7-12, which are optional and not necessary components of “capture probes.” Thus, molecules that include a nucleic acid

²⁶ Gerry et al., Universal DNA Microarray Method for Multiplex Detection of Low Abundance Point Mutations,” 292 *J. Mol. Biol.*, 251–262 (1999). (Ex. 2028).

binding site but lack any type of label – such as PCR primers – must fall within the scope of a “capture probe” as set forth in the Specification. Ex1225 ¶¶38–39.

Reply 11; *see also* Ex. 1225 ¶ 39 (Dr. Quackenbush’s reasoning that “[t]he specification is explicit that biotin and streptavidin are considered ‘linkers’ or labels,” ’394 patent, 14:22–46, 23:23–36, which are optional and not necessary components of ‘capture probes.’ Thus, molecules that include a nucleic acid binding site but lack any type of additional chemical moiety (label or linker) such as biotin, must fall within the scope of a ‘capture probe’ as set forth in the Specification. This includes PCR primers, which are nucleic acids that hybridize to a target (i.e., contain a nucleic acid binding site), but which do not necessarily comprise linkers and/or labels.”)

Although the issued claims of the ’394 patent recite “capture probes” rather than the “pull-down probes” limitation sought earlier in prosecution (*see* DI 29 (discussing Ex. 1078, 3; 2004, 17)), Patent Owner’s proposed construction requires that “capture probes” “*physically capture and isolate the target DNA*” (POR 17), and “exclusively . . . refer to hybridization capture” (Sur-reply 2). This construction is contrary to the Specification for the reasons set forth in paragraphs 43–51 of Dr. Quackenbush’s Second Declaration, which we adopt, and as further set forth below. In short, while the scope of “capture probe” set forth in the Specification clearly encompasses a biotinylated oligonucleotide bait and a magnetic bead system, which we agree may be used to physically capture and isolate a target DNA, Patent Owner chose not to describe or claim its invention so narrowly.²⁷ To the contrary, the Specification discloses that the claimed

²⁷ Patent Owner cites *TIP Systems* for the proposition that “the claims of the patent need not encompass all disclosed embodiments. Our precedent is

“capture probes” may instead comprise a vast assortment of “linkers” and “labels” that suggest neither physical capture nor isolation of a nucleic acid, including nucleotides, radioisotopes, fluorophores, chemiluminophores, chromophores, electron donors/acceptors, dyes, spin labels, and even mere “change(s) in mass” detectable by surface plasmon resonance. *See, e.g.,* Ex. 1001, 14:22–47, 31:4–12, 41:32–42:37, 42:49–51 (“Suitable linkers comprise any chemical or biological compound capable of attaching to a label, primer, and/or capture probe disclosed herein.”). Consistent with the lack of any requirement for either physical capture or isolation, or indeed, “target DNA,” Petitioner correctly points out that the relevant limitations of claim element 1(c) require only “contacting” a nucleic acid sample with a capture probe, whereas the Specification teaches that any subsequent “separation” is an “alternative[] or additional[]” step. *See* Reply 12–13 (citing Ex. 1001, 32:12–18, 26–33, 37:5–8; Ex. 1225 ¶¶ 42–47). As further discussed below, we read neither “capture probes,” nor the structure of the asserted claims to require a physical isolation.²⁸ *See* Tr. 26:12–21 (Patent Owner’s counsel arguing that “physical isolation by pullout is implicitly

replete with examples of subject matter that is included in the specification, but is not claimed.” POR 21 (citing *TIP Systems, LLC v. Phillips & Brooks/Gladwin Inc.*, 529 F.3d 1364, 1373 (Fed. Cir. 2008)). But the present case represents the converse—where, the claims do not limit the scope of a term so broadly described in the Specification.

²⁸ Patent Owner contends that “capture probes can be physically separated from [solid support] gels by cutting out the strip of nitrocellulose containing the captured nucleotides of interest and eluting the captured nucleotides.” POR 21 (citing Ex. 2031 ¶65; Ex. 2026). But even if “capture probes” and/or claim element 1(d) were to suggest a separation step, it was well known that PCR products could be similarly isolated, which further renders Patent Owner’s argument unavailing. *See* Ex. 1225 ¶¶ 84–88 (citations omitted).

required to do the sequencing of limitation 1(d)"); *see generally id.* at 26:12–28:9.

Turning to extrinsic evidence, Patent Owner argues that one of ordinary skill in the art would recognize that PCR requires *two* primers (a primer pair) to amplify DNA, whereas using, e.g., biotinylated oligonucleotide baits and a magnetic bead system, “only *one* capture probe is required to physically pulldown a target DNA strand.” POR 21–22 (citing Ex. 1020 ¶¶ 72–76; Ex. 2018, 112:18–115:11; Ex. 2031 ¶ 66²⁹). But while the basic methodology of using PCR primers and biotinylated oligonucleotide baits are not in dispute, we find nothing in the Specification suggesting that a “capture probe” cannot encompass individual PCR primers and PCR primer pairs. *See e.g.*, Ex. 1001, 32:12–15 (“one or more subsets of nucleic acid molecules may be produced by *contacting a nucleic acid sample with one or more* beads, *capture probes*, labels, or a combination thereof”) (emphasis added); *id.* at 41:32–33 (“The capture probes, primers, labels, and/or beads may comprise one or more nucleotides.”). Indeed, the Specification’s teaching that “[t]he methods disclosed herein may comprise 1 or more, 2 or more . . . capture probes or capture probe sets. The one or more capture probes or capture probe sets may be different, similar, identical, or a combination thereof,” indicates that “capture probe” should be read to encompass *both* individual PCR primers (where the one or more

²⁹ Dr. Furneaux’s assertion that this indicates “physical differences between PCR primers and capture probes” is not persuasive insofar as nothing in the Specification prohibits a capture probe from comprising a primer or primer pair. Moreover, claim steps 1(c) and 1(e) refer to a “plurality of capture probes,” which encompass one or more primers and/or primer pairs.

capture probes are identical), and PCR primer pairs (where the one or more capture probes are different). *See id.* at 31:13–23.

We are also not persuaded by Dr. Furneaux’ s contention that “capture probes are uniquely different from PCR in that capture probes physically isolate native DNA.” *See* Ex. 2031 ¶ 74; *see also* Tr. 25:24–26:9 (Patent Owner’s counsel emphasizing that capture probes must isolate original target DNA rather than copies thereof). First, we credit Dr. Quackenbush’s testimony that this statement is not correct on its face because “[s]olution-phase hybridization capture baits are frequently used in experiments where the target DNA has been PCR amplified and modified with sequencing adaptors before capture.” *See* Ex. 1225 ¶ 50 (citing Ex. 1022, 91; Ex. 2031³⁰ ¶ 62). Moreover, as stated by Petitioner, “both PCR primers and solution-phase baits are used to preserve sequence information and generate enough DNA for subsequent sequencing.” Reply 13 (citing Ex. 1225 ¶¶ 49–51); *see also* Ex. 1020 ¶¶ 66–78.

We further credit Petitioner’s evidence that persons of ordinary skill in the art understood PCR as a sequence capture technique. *See* Reply 14–15 (citing Ex. 1222, 95:15–18, 97:4–21, 170:21–24, Ex. 1030, 3 (PCR primer sets used “to capture the starting molecules present in the template”), Ex. 1216 (“The major differences with all the platforms are the way in which multiple loci are captured and amplified. Microfluidics, emulsion PCR and oligo-probes are the different ‘capture’ mechanisms”); Ex. 1217, 3-4 (“There are two basic approaches to sequence capture: PCR-based and

³⁰ Gnirke *et al.*, “Solution hybrid selection with ultra-long oligonucleotides for massively parallel targeted sequencing,” 27(2) *Nature Biotechnology* 182–189 (2009). Ex. 1005.

hybridization-based, the latter of which can be done either in solution or on the surface of an oligonucleotide microarray.” (further referencing “more traditional PCR-based capture”); Ex. 1225 ¶¶ 54–57).

Relatedly, Patent Owner contends that one of ordinary skill in the art would understand that PCR primers have different functions and are categorized according certain “core reaction principle[s].” POR 22–23.

According to Patent Owner,

the core reaction principle of capture probes involves hybridizing to “targeted regions of interest, either in solution or on a solid support, *so that one can physically capture and isolate the sequences of interest.*” *Id.* In contrast, the core reaction principle of PCR primers is “*amplification*” of targeted regions by a “polymerase chain reaction (PCR)” that *copies* target regions. *Id.*; *see also id.* 376, Fig. 1.

Id. at 22–23 (further citing Ex. 1020 ¶ 70–75; Ex. 2158, 2650; Ex. 2157, 320, Ex. 2157, 320; Ex. 2031 ¶¶ 67–68). Patent Owner similarly argues that one of ordinary skill in the art would select a particular enrichment procedure (e.g., PCR, on-array hybrid capture, in-solution hybrid capture) best suited to a particular research project or field of application. *See id.* at 24–27 (citing e.g., Ex. 1006, 375, 378–80; Ex. 1006, 383, Ex. 1007, 114, Table 1; Ex. 2031 ¶¶ 71–74; Sur-reply 8–9. For the reasons set forth in paragraphs 52 and 53 of Dr. Quackenbush’s Second Declaration, which we adopt, we do not find Patent Owner’s argument persuasive. *See* Ex. 1225 ¶¶ 52–53 (citations omitted). In short, that a person of ordinary skill in the art would evaluate the pros and cons of available enrichment techniques says little about whether PCR primers are within the scope of “capture probes” as described in the instant Specification.

For the reasons set forth above, and in light of all the evidence of record we construe “capture probes,” as used in the instant Specification, to

encompass PCR primers and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits).

D. Ground 1: Obviousness in View of Leary

As Ground 1, Petitioner challenges claims 1–11 and 13–19 as unpatentable for obviousness in view of Leary. Pet. 18. Supported by the testimony of Dr. Quackenbush, Petitioner provides an element-by-element comparison of the claims to the cited art. *See id.* at 28–54; Reply 1–17; Resp. 1–3; Ex. 1020 ¶¶ 109–209; Ex. 1225 ¶¶ 58–103.

Patent Owner opposes. *See* POR 5–36; Sur-reply 2–19; Ex. 2007 ¶¶ 12–35; Ex. 2031 ¶¶ 84–154. We address with particularity only those elements raised by Patent Owner, finding the remaining elements supported by Petitioner’s un rebutted arguments and evidence. As Leary is discussed above in Section II.C.1.b.i, we begin with Patent Owner’s arguments.

1. “whole genome sequencing”

Patent Owner contends that “Leary does not teach or suggest the ‘whole genome sequencing’ (WGS) required by limitation 1(b).” POR 5. For the reasons set forth in section II.C.1, above, Petitioner has sufficiently shown that Leary discloses or renders obvious this element. In, particular, we construe whole genome sequencing as “a non-targeted approach to sequencing all or almost all of a genome,” which encompasses a non-targeted approach to sequencing about 50% or more of a genome. We further determine that the application of the mate-paired tag/PARE technique in Leary satisfies this definition as it is a non-targeted approach, which results in the sequencing of about 53% or 60% of the genome (depending on the calculation applied). *See* Reply 6; Sur-reply 14; Ex. 1225 ¶ 66. Moreover, the art of record evidences that one of ordinary skill in the

art at the time of the invention recognized that Leary discloses whole genome sequencing.³¹

2. “capture probes”

Patent Owner contends that Leary fails to disclose “capture probes” as used in limitations 1(c)i–ii, when construed in the way urged by Patent Owner. POR 16. In section II.C.2, above, we construe “capture probes,” as used in the instant Specification, to encompass PCR primers and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits). Because Leary discloses the use of PCR primers, Petitioner has sufficiently shown that Leary discloses “capture probes,” as would have been understood by one of ordinary skill reading the ’394 Specification.³²

3. “plurality of capture probes”

Patent Owner further contends that Leary fails to teach the “plurality of capture probes” recited in limitation 1(e). POR 31–33; Sur-reply 18. In addition to its contention that Leary does not teach capture probes at all (discussed above), Patent Owner points to Petitioner’s reliance on Leary’s Figure 4 for a “plurality of capture probes.” *Id.* at 31. Patent Owner contends

³¹ Given that Leary discloses “whole genome sequencing,” properly construed, we need not address Patent Owner’s argument that one of ordinary skill in the art “would need to modify Leary to . . . cover all the base pairs in the human genome.” Sur-reply 14–16; *see also* Section II.C.1.a, above (finding that sequencing *all* of a genome was unachievable at the time of the invention).

³² Given that Leary discloses “capture probes,” properly construed, we need not address Patent Owner’s argument that one of ordinary skill in the art “would not have been motivated to completely redesign Leary’s PCR primer-based targeted PARE assay to [for example,] . . . change the core reaction principle from PCR primers to capture probes.” *See* POR 34.

that this disclosure relates to the analysis of “a single chromosomal rearrangement . . . with a single primer pair.” *Id.* (citing Ex. 2031 ¶¶ 143–145). Patent Owner similarly argues that although Leary Figure 1 indicates the preparation of multiple mate-pair tags to identify “patient-specific rearrangements,” “the actual experiment conducted by Leary . . . monitored just one chromosomal rearrangement”—also presumably using a single primer pair. *See id.* at 32 (citing Pet. 38–30). As such, Patent Owner argues that Petitioner “fails to explain how Leary’s single primer pair that function(s) as a single unit constitutes a plurality of capture probes.” *Id.* at 32 (citing Ex. 2018 114:21–115:11; Ex. 2160; Ex. 1006, 380).

We do not find Patent Owner’s argument persuasive for the reasons set forth on page 16 of the Reply, which we adopt. *See* Reply 16 (citing Ex. 1225 ¶¶ 89–92). As noted in section II.C.2, above, the breadth of the Specification encompasses individual PCR primers and PCR primer pairs within the scope of “capture probes.” *See* Ex. 1001, 31:13–23 (“The methods disclosed herein may comprise 1 or more, 2 or more . . . capture probes or capture probe sets. The one or more capture probes or capture probe sets may be different, similar, identical, or a combination thereof.”). In practice, each individual primer of a primer pair will be present in many copies such that both the individual primer, and a single pair of primers satisfy the “multiplicity” requirement. To the extent Patent Owner implicitly reads “a plurality of capture probes” as requiring different nucleic acid sequences that bind to different targets, such a “requirement” is met by the multiplicity of copies of a single PCR primer pair, which Patent Owner admits is disclosed in Leary’s Figure 4.

In the alternative, even were we to accept Patent Owner’s argument that the “plurality of capture probes” limitation is not satisfied by a *single* PCR primer pair, we credit Dr. Quackenbush’s testimony that Leary “expressly motivates tracking multiple PCR amplicon biomarkers in the same patient,” and implicitly describes or renders obvious the use of multiple PCR primer pairs as required under claim step 1(e). *See* Ex. 1225, ¶ 92.

For the reasons set forth above, Petitioner has sufficiently shown that Leary discloses “a plurality of capture probes,” as would have been understood by one of ordinary skill reading the ’394 Specification.

4. “second nucleic acid sample”

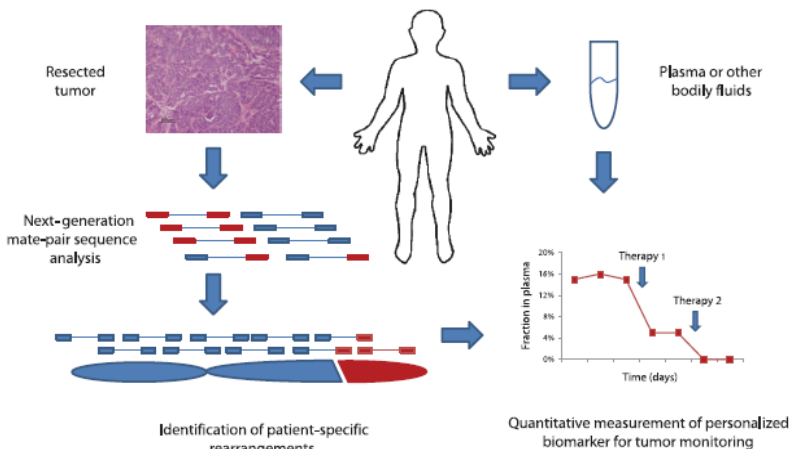
Claim step 1(c) recites “generating a second subset of nucleic acid molecules from a second nucleic acid sample of said individual... wherein said first sample or said second sample is obtained from said subject at a first time point.” With respect to this limitation, the Petition states that “Leary obtains tumor samples at a first time point, when the tumor sample is resected, and also discloses obtaining plasma samples at multiple time points after tumor resection.” Pet. 36–37 (citing Ex. 1002, 1, 2 (Fig 1), 4; Ex. 1020 ¶¶ 136–137) (emphasis removed).

Patent Owner contends that because Petitioner relies on the initial resected tumor sample for both limitations 1(a) and 1(c)(1), that it has failed to identify a *second* nucleic acid sample in Leary. POR 15–16. With respect to this aspect of its proofs, Petitioner contends that one of ordinary skill in the art would understand that different nucleic acid samples can be taken from the same tumor sample. Reply 9–10 (citing Ex. 1020 ¶ 99). Patent Owner reasonably responds that Petitioner has not established *why* the

skilled artisan would extract a second nucleic sample from the initial resected tumor sample. Sur-reply 17. Patent Owner's argument is ultimately unavailing as it mis-understands the argument in the Petition. *See* Pet. 36–37 (citing Ex. 1020 ¶¶ 136–137).

In particular, Petitioner and Dr. Quackenbush point to Leary's Figure 1, reproduced below, as illustrating Leary's disclosure of first and second nucleic acid samples. Pet. 37; Ex. 1020 ¶ 137.

Fig. 1. Schematic of PARE approach. The method is based on next-generation mate-paired analysis of resected tumor DNA to identify individualized tumor-specific rearrangements. Such alterations are used to develop PCR-based quantitative analyses for personalized tumor monitoring of plasma samples or other bodily fluids.



Leary's Figure 1 is a schematic of the PARE approach, illustrating that a first DNA sample is taken from a patient's resected tumor, and a second DNA sample is taken from the same patient for "tumor monitoring of plasma samples or other bodily fluids." Ex. 1002, Fig. 1. The graph in the lower right-hand corner of Figure 1 further illustrates a series of seven samples taken from the same patient to monitor tumor load in the patient over time. *Id.*; *see also* Ex. 1020 ¶¶ 143–146 (further discussing the detection of a patient's tumor DNA in successive samples in Leary's Figures 1 and 4).

We also credit Dr. Quackenbush's testimony that the text of Leary similarly "discloses that the second sample may be a plasma sample from the patient taken at time points after the initial tumor resection." Ex. 1020

¶ 137. In particular, Leary states:

To determine whether the rearranged sequences could actually be detected in clinical samples, we evaluated circulating DNA from plasma samples of patients Hx402 and Hx403. ***The sample from patient Hx403 was obtained before surgery, whereas the samples from patient Hx402 were obtained before and after surgery.*** . . . PCR amplification of plasma DNA using primers spanning the breakpoints produced products of the expected sizes only in the plasma samples from patients with disease and not in plasma from healthy controls (Fig. 4A). Sequencing the PCR products from plasma DNA identified the identical breakpoints observed in the tumor DNA samples.

Id. (quoting Ex. 1002, 4).

In view of the above, Petitioner has sufficiently shown that Leary discloses the “second nucleic acid sample” recited in claim step 1(c).

5. “third sequencing assay” and Claim 19

Claim step 1(f) recites, in part, “***conducting a third sequencing assay*** on said additional subset of nucleic acid molecules from said additional nucleic acid sample to yield a third result comprising a third nucleic acid sequence.” Claim 19 similarly recites “repeating said third sequencing assay one or more times” on “a fourth nucleic acid sequence.” As we understand the Petition, Petitioner and Dr. Quackenbush point to Leary’s PCR amplification and monitoring of successive plasma samples over time as shown in Figures 1 and 4 as evidencing a third (and additional) subsets of nucleic acid molecules. *See* Pet. 41; Ex. 2020 ¶¶ 147–148. Petitioner then points to Leary’s statement regarding the “[s]equencing of the PCR products from plasma DNA’ to confirm whether the rearrangement observed in the plasma ‘identified the identical breakpoints observed in the tumor DNA samples.’” Pet. 41. “Accordingly,” Petitioner contends, “Leary discloses, and at a minimum renders obvious, conducting a sequencing assay

on PCR products generated from each plasma sample, including the plasma sample from which the third subset of nucleic acid molecules was generated.” *Id.* (citing Ex. 2020 ¶ 148). Petitioner further contends that claim 19 would similarly be obvious because “Leary discloses reanalyzing the patient’s biomedical status (e.g., cancer progression) at multiple time points (at least six) following the first sequencing assay, including by monitoring tumor-associated nucleic acid levels in multiple samples collected at different subsequent timepoints and plotting them on a biomedical report to monitor that status.” Pet. 53–54 (citing Ex. 1002, Fig. 4B, Ex. 1020 ¶ 207–209).

But as Patent Owner points out, Petitioner relies on Leary’s disclosure of sequencing PCR products from plasma DNA to confirm whether the rearrangements observed in the plasma “identified the identical breakpoints observed in the tumor DNA samples,” as covering the “second sequencing assay on said second subset of nucleic acid molecules,” recited in claim step 1(d). POR 33 (citing Pet. 37–38). As we understand the reference, Leary sequenced PCR products obtained from the initial plasma DNA samples, but did not conduct this sequencing at later time points. *See* Ex. 1004, 4. As such, we agree with Dr. Furneaux that “Leary does not disclose conducting an additional verification (i.e., a re-verification) sequencing step,” i.e., the “third sequencing assay” recited in claim step 1(f), nor the additional sequencing steps required by claim 19. *See* Ex. 2031 ¶¶ 147, 154.

With respect whether it would have been obvious to conduct further sequencing on Leary’s later plasma samples, Patent Owner and Dr. Furneaux contend that such re-verification sequencing was unnecessary,

and “Leary’s workflow was already ‘difficult to implement on a routine basis’ and more sequencing steps would be expected to make the workflow even more difficult to implement and lead to even less adoption. Adding sequencing steps to Leary’s workflow would add more data analysis, cost, and time.” Ex. 2031 ¶¶ 147–148, 154 (citing, e.g., Ex. 1030, 8, 10); POR 33–34.

We do not find Patent Owner’s argument with respect to obviousness persuasive for the reasons in the Petition, which we adopt. *See* Pet. 41–44 (citing, e.g., Ex. 1020 ¶¶ 148–154). Briefly, and summarized in the Reply, one of ordinary skill in the art would have “recognized that performing additional sequencing had become, and was becoming, significantly cheaper and more accessible between the publication of Leary and the priority date.” Reply 17 (citing Ex. 1020 ¶ 151; Ex. 1222, 128:23–129:19). Indeed, Dr. Furneaux testified that, even though “Leary’s workflow only sequences the tumor and matched normal sample once,” Leary reports “seven cases where the amplicons were sequenced for verification” (albeit without reporting the sequence results)—which tends to undercut Patent Owner’s argument regarding the burden of additional sequencing even at the time Leary was published. *See* Ex. 2031 ¶ 148.

Against this backdrop of increasing availability of sequencing information, Petitioner plausibly asserts that additional sequencing of “Leary’s PCR products would have been motivated at least by the desire of cancer patients to obtain the most complete and accurate information available, given that sequencing PCR products involved no more than routine skill.” Reply 16 (citing Ex. 1020 ¶¶ 150, 153 (noting, for example, that such an approach was employed by Forshew)). And as further

articulated by Dr. Quackenbush, one of ordinary skill in the art “would have been particularly motivated to use a confirmatory sequencing step in the context of Leary’s methods, given that the purpose of the analysis is to predict the status of the patient’s cancer.” Ex. 1020 ¶ 150. And “[p]roviding the inquisitive patient and their physician with further information regarding the progress of the patient’s treatment would . . . have further motivated a POSA to conduct sequencing of the nucleic acid subsets generated at additional timepoints.” *Id.* (citing Ex. 1002, 8).

On balance, and for the reasons noted above, Petitioner has sufficiently shown that Leary renders obvious the “third sequencing assay” recited in claim step 1(f), as well as the subsequent step of “repeating said third sequencing assay one or more times,” required by dependent claim 19.

6. Claims 17 and 18

Dependent claims 17 and 18 require that the “whole genome sequencing produces at least [750,000,000 or 1,000,000,000] sequencing reads.” According to Petitioner, “[d]etermining the optimal number of sequencing reads for a given application would have required no more than routine optimization of predictable and well-known sequencing techniques,” and a person of ordinary skill in the art “would have been motivated to sequence to greater depths of coverage in the context of cancer detection and monitoring to increase the accuracy and reduce the error rate of the assay.” Pet. 52; *see also* Ex. 1020 ¶¶ 201–206.

Petitioner contends that such an increase in coverage “would have been obvious to implement, especially given that sequencing times and costs decreased rapidly between 2010 and 2013.” *Id.* According to Dr. Quackenbush, Leary discloses that its whole genome sequencing was to

“an 18-fold physical coverage of the human genome” whereas “sequencing to greater depths of coverage was routine by 2013, in particular using the Illumina sequencers that had become the industry standard,” and sequencing the claimed number of reads, “involves, at most, repeating experiments that were already performed successfully.” Ex. 1020 ¶¶ 203–204 (citing Ex. 1002, 1; Ex. 1028 (“[I]t is very easy to increase the coverage or sequence depth, if you later decide you need more data.”); *see also* Ex. 2018, 94:8–97:4 (Dr. Quackenbush’s testimony that sequence coverage depends on the application)).

In illustrating that it would have been routine to achieve the claimed number of sequencing reads, Dr. Quackenbush calculates that “using *Leary’s* 25 bp read length and sequencing to a depth of coverage of 30x would have resulted in 3,600,000,000 sequencing reads.” Pet. 52–53 (citing Ex. 1020 ¶¶ 201–206). Patent Owner responds that, without more, the selection of 30x sequencing depth “reveals improper hindsight” and that Petitioner provides no reasoned analysis why a person of ordinary skill would increase coverage to 30x as opposed to any other multiplier. POR 35 (citing, e.g., Ex. 2031 ¶¶ 151–152).

On balance, Petitioner has the better argument. In sum, we agree with Petitioner’s argument and evidence that one of ordinary skill in the art would have been motivated to increase sequence coverage to increase the accuracy and reduce the error rate of the assay, and that achieving the number of sequence reads recited in claims 17 and 18 would have required no more than routine optimizations of predictable and well-known sequencing techniques.

Accordingly, Petitioner has shown sufficiently that dependent claims 17 and 18 are obvious in view of Leary.

7. Objective Indicia of Non-Obviousness

Objective indicia of non-obviousness may include long-felt but unsolved need, failure of others, unexpected results, commercial success, copying, licensing, industry praise, and expert skepticism. *Mintz v. Dietz & Watson, Inc.*, 679 F.3d 1372, 1379 (Fed. Cir. 2012). “[O]bjective indicia ‘may often be the most probative and cogent evidence of nonobviousness in the record,’” and “help turn back the clock and place the claims in the context that led to their invention.” *Id.* at 1378. Evidence of objective indicia of non-obviousness “must always when present be considered en route to a determination of obviousness.” *Transocean Offshore Deepwater Drilling, Inc. v. Maersk Drilling USA, Inc.*, 699 F.3d 1340, 1349 (Fed. Cir. 2012); *see also Apple Inc. v. Samsung Elecs. Co.*, 839 F.3d 1034, 1048 (Fed. Cir. 2016) (en banc). Such evidence, however, does not necessarily control the obviousness conclusion. *See, e.g., Pfizer, Inc. v. Apotex, Inc.* 480 F.3d 1348, 1372 (Fed. Cir. 2007). (“Here, the record establishes such a strong case of obviousness that Pfizer’s alleged unexpectedly superior results are ultimately insufficient.”)

But objective indicia of non-obviousness are “only relevant to the obviousness inquiry ‘if there is a nexus between the claimed invention and the [objective indicia of non-obviousness].’” *In re Affinity Labs of Tex., LLC*, 856 F.3d 883, 901 (Fed. Cir. 2017) (quoting *Ormco Corp. v. Align Tech., Inc.*, 463 F.3d 1299, 1312 (Fed. Cir. 2006)). For objective indicia of non-obviousness to be accorded substantial weight, their proponent must establish a nexus between the evidence and the merits of the claimed

invention. *ClassCo, Inc. v. Apple, Inc.*, 838 F.3d 1214, 1220 (Fed. Cir. 2016). “[A] patentee is entitled to a rebuttable presumption of nexus between the asserted evidence of secondary considerations and a patent claim if the patentee shows that the asserted evidence is tied to a specific product and that the product ‘is the invention disclosed and claimed.’” *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (quoting *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1392 (Fed. Cir. 1988)).

Relying on paragraph 178 of Dr. Furneaux’s Second Declaration (Ex. 2031), Patent Owner contends it is “entitled to a presumption of nexus because Personalis’ NeXT Personal[®] assay embodies claim 1.” POR 81; Sur-reply 26. Patent Owner further asserts—without citation or elaboration—that “the evidence independently shows a nexus.” POR 81. Patent Owner further contends that its Personalis’ NeXT Personal assay has “received industry praise due in part to its sensitivity,” which it contrasts to that of Leary (POR 81–82 (citing Ex. 1002, 4, Fig. 4B; Ex. 2001; Ex. 2035 ¶ 11, 14–16, 18–20; Ex. 2153; Ex. 2154; Ex. 2156)), and argues that “industry[] experts held skepticism toward solving the challenges of conducting highly sensitive genome sequencing using capture probes” and “how to handle [a] massive amount of sequencing data” (*id.* at 83–84 (citing Ex. 1030, 10; Ex. 1031, 686; Ex. 1004, 48; Ex. 2031 ¶¶ 181–182; Ex. 1017, 3)).

Paragraph 178 of Dr. Furneaux’s Second Declaration, the only substantive evidence it provides in support of nexus, is a bare statement that “Personalis is entitled to a presumption of nexus because, as detailed in the following chart, Personalis’ NeXT Personal[®] assay embodies claim 1 of the

'394 Patent.” Ex. 2031 ¶ 178. The referenced chart appears to compare claim 1 of the '394 patent to cut-and-paste excerpts of Exhibit 2012 (Personalis NeXT Personal Brochure). *Id.*

In our Institution Decision, we considered Petitioner’s argument that “Patent Owner’s nexus evidence is supported by ‘only a cursory attorney-created claim chart,’” and invited the parties to more fully “present argument and evidence on this issue,” such that we could “consider objective evidence of non-obviousness after development of a full trial record.” DI 33. Patent Owner’s mere inclusion of the same claim chart in Dr. Furneaux’s Second Declaration—absent any substantive investigation or guidance by Dr. Furneaux—and, moreover, absent any substantive explanation or argument in Patent Owner’s briefs, does not suffice. *See* Ex. 1223, 35:15–38:24 (Dr. Furneaux’s testimony that his input in reviewing the claim chart was merely “to check that what’s in the left-hand-side column there is, in fact, recapitulated in the right-hand-side column.”). Patent Owner has thus, “left it up to us to figure out” whether nexus exists. *See* Reply 27 (citing *Microsoft Corp. v. Syncloud Technologies, LLC*, No. 2020-00316 (P.T.A.B. Jun. 14, 2021)). We decline to do so. *See DeSilva v. DiLeonardi*, 181 F.3d 865, 867 (7th Cir. 1999) (“A brief must make all arguments accessible to the judges, rather than ask them to play archaeologist with the record.”). Considering the still incomplete record before us, Patent Owner has not satisfied its threshold burden of showing that its asserted evidence of secondary considerations is tied to a specific product, where that product “is the invention disclosed and claimed.” *See Fox Factory*, 944 F.3d at 1373.

A *prima facie* case of nexus cannot be made by simply showing that “the patent claims broadly cover the product that is the subject of the

evidence of secondary considerations.” *See Fox Factory*, 944 F.3d at 1377. Accordingly, even were we to accept Dr. Furneaux’s conclusory testimony that claim 1 “embodies” (i.e., covers) the NeXT Personal assay, that is not alone sufficient to establish coextensiveness and the necessary degree of correspondence between the product and scope of the claims entitling Patent Owner here to a presumption of nexus. Dr. Furneaux does not, for example, explain why alleged success of the NeXT Personal assay is even substantially due to “the invention disclosed and claimed” as opposed to, for example, some significant unclaimed features. *See id.* at 944 F.3d at 1373.

In view of the above, we find that Patent Owner has not met its burden on the nexus issue. *WMS Gaming Inc. v. Int’l Game Tech.*, 184 F.3d 1339, 1359 (Fed. Cir. 1999) (“The patentee bears the burden of showing that a nexus exists.”).

We further note that, “[w]here the offered secondary consideration actually results from something other than what is both claimed and *novel* in the claim, there is no nexus to the merits of the claimed invention.” *In re Kao*, 639 F.3d 1057 (Fed. Cir. 2011). In this respect, Patent Owner anchors its evidence of industry praise and skepticism largely on the alleged sensitivity of its commercial product. POR 83–84 (e.g., referencing “the challenges of conducting highly sensitive genome sequencing using capture probes” and “difficulties regarding ‘how to handle the massive amount of sequencing data’”). But the sensitivity allegedly displayed by Patent Owner’s commercial product is not claimed, nor even discussed in the Specification. Nor does Patent Owner link any patent claim or portion of the Specification to a solution for handl[ing] . . . “massive amount(s) of sequencing data.” *See* POR 84 (citing Ex. 1017, 3). Moreover, to the extent

Patent Owner's alleged high sensitivity derives from the use of "genome sequencing using capture probes," such techniques would appear to be encompassed by the prior art under our construction of "capture probes."

Considering the above, we ascribe no weight to Patent Owner's secondary considerations evidence.

8. Conclusion as to Ground 1, Claims 1–11 and 13–19

Considering the record before us, and in light of our constructions of "whole genome sequencing" and "capture probes," we find that Petitioner has shown persuasively that claims 1–11, 13–19 are unpatentable in view of Leary.

E. Ground 2: Obviousness in View of Leary and Ley

As to Ground 2, Petitioner further challenges claims 1–11 and 13–19 as unpatentable for obviousness in view of Leary, as discussed above, and further in view of Ley. *See* Pet. 11, 54–58; Reply 17–18; Ex. 1020 ¶¶ 210–220; Ex. 1225 ¶¶ 104–107. Patent Owner opposes. *See* POR 36–37; Sur-reply 19–20; Ex. 2031 ¶¶ 155–162. As Leary is discussed above, we begin with an overview of Ley.

1. Overview of Ley (Exhibit 1012)

Ley discloses the use of whole-genome sequencing to identify tumor-specific single nucleotide variants (SNVs) in a patient with acute myeloid leukemia. *See, e.g.*, Ex. 1012, Abstract, 67. According to Ley:

Whole-genome sequencing is now possible at a reasonable cost and timeframe to use this approach for the unbiased discovery of tumour-specific somatic mutations that alter the protein-coding genes. Here we present the results obtained from sequencing a typical acute myeloid leukaemia genome, and its matched normal counterpart obtained from the same patient's

skin. We discovered ten genes with acquired mutations; two were previously described mutations that are thought to contribute to tumour progression, and eight were new mutations present in virtually all tumour cells at presentation and relapse.

Id. at Abstract. In particular, Ley “used massively parallel sequencing technology to sequence the genomic DNA of tumour and normal skin cells obtained from a patient with a typical presentation of French-American-British (FAB) subtype M1 acute myeloid leukaemia (AML) with normal cytogenetics.” *Id.* at 66.

Using a commercially-available Illumina Genome Analyzer, Ley achieved about 82% and 90% diploid coverage of the tumor and normal skin cell genomes, respectively. *Id.* at 67. Ley also reports sequencing data comprising “32.7-fold ‘haploid’ coverage (98 billion bases)” for the tumor genome, “and 13.9-fold coverage (41.8 billion bases) . . . for the normal skin sample.” *Id.* at 66. According to Ley:

After alignment to the human reference genome, sequence variants were discovered in the tumour genome and compared to the patient’s normal sequence, to the dbSNP database, and to variants recently reported for two other human genomes; revealing new single nucleotide and small insertion/deletion (indel) variants genome-wide. Somatic mutations were detected in genes not previously implicated in AML pathogenesis, demonstrating the need for unbiased whole-genome approaches to discover all mutations associated with cancer pathogenesis.

Id. at 67 (internal endnote numbers omitted). Ley further reports that “[f]or the purposes of this initial study, we restricted our downstream analysis to the coding sequences of annotated genes,” noting that these included alterations in eight protein-coding sequences including “previously described, recurrent somatic insertions in the *FLT3* and *NPM1* genes.” *Id.* at 66.

Using a panel of 10 tumor-specific mutations, Ley compared the initial sequencing data to that from the same patient after relapse:

To better define the percentage of tumour cells that contained each of the discovered somatic mutations, we amplified each mutation containing locus from non-amplified genomic DNA derived from the *de novo* and relapse tumour samples, and from the skin biopsy obtained at presentation. The resulting amplicons were sequenced using the Roche/454 FLX platform, and the frequency of reads containing the reference and variant alleles were defined (Fig. 3 and Table 3).

Id. at 69; *see also id.* (Figure 3 legend noting that “the normal skin sample was contaminated with leukaemic cells containing the somatic mutations. The patient’s white blood cell count was 105,000 (85% blasts) when the skin punch biopsy was obtained.”).

As indicated above, Table 3, reproduced below, shows the frequency of reads containing the reference and variant alleles.

Table 3 | 454 Readcount data for somatic mutations and known SNPs

Gene	Consequence	Primary AML (100% blasts)			Skin			Relapse (78% blasts)		
		Variant	Ref	Variant (%)	Variant	Ref	Variant (%)	Variant	Ref	Variant (%)
CDH24	Y590X	5672	4890	53.70	564	10358	5.16	3108	4599	40.33
SLC15A1	W77X	3817	4962	43.48	875	10773	7.51	4714	7173	39.66
KND1	L799F	4640	4848	48.90	770	8972	7.90	3883	6342	37.98
PTPR	P1235L	998	1058	48.54	126	1489	7.80	350	493	41.52
GRINL18	R176H	2211	2674	45.26	318	4461	6.65	1447	2070	41.14
GPR123	T38I	4618	4569	50.27	850	9751	8.02	3660	6057	37.67
EBI2	A338V	12750	15453	45.21	458	10088	4.34	2646	3627	42.18
PCLKC	P1004L	992	855	53.71	341	3153	9.76	705	773	47.70
FLT3	ITD	4220	7810	35.08	3475	23159	13.05	3870	8495	31.30
NPM1	CATG ins	1550	1974	43.98	143	2390	5.65	2303	3910	37.07
BRCA2	N372H	778	752	50.85	763	876	46.55	285	303	48.47
TP53	P72R	8989	1	99.99	8161	0	100.00	7914	6	99.92

The differences between variant frequencies in primary or relapse tumour samples and skin were highly significant for all somatic mutations ($P < 0.000001$, Fisher’s exact test, one tailed). The BRCA2 variant is a known heterozygous SNP in this genome, and the TP53 variant is a known homozygous SNP.

Table 3 shows Roche/454 FLX readcount data for ten tumor-specific mutations and two known SNPs (BRCA2 and TP53) in the primary tumour, relapse tumour, and skin specimens. *Id.*

Figure 3 of Ley is reproduced below.

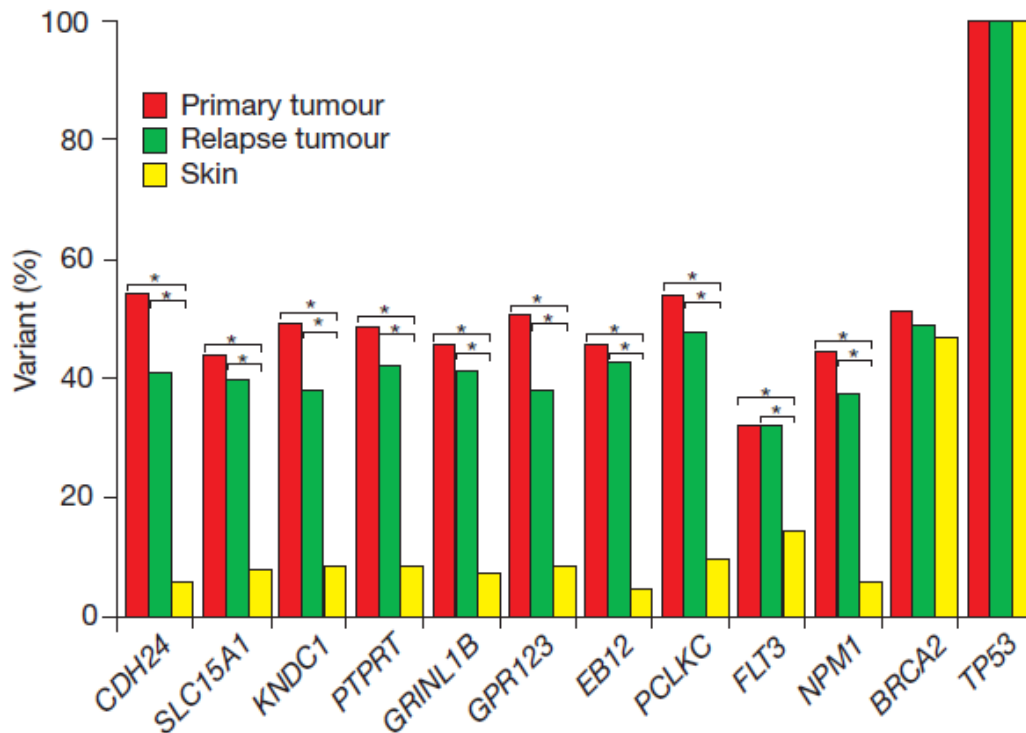


Figure 3 is a graphical representation the data shown in Table 3, above, where the asterisks indicate statistical significance as compared to the skin sample for all mutations ($P < 0.000001$, Fisher's exact test).

2. Analysis

In short, Petitioner contends that the addition of “Ley further renders obvious alternative embodiments within the scope of the claim step 1(c), including selecting different types of polymorphisms as well as polymorphisms ‘based on or extracted from one or more databases of polymorphisms’ whether alone or in combination with the polymorphisms Leary identified in its samples.” Pet. 54–55 (emphasis omitted). As described by Petitioner, Ley detected these polymorphisms via whole-genome sequencing, and then used probe-based capture techniques like Leary’s to “monitor[] polymorphisms that were found in ‘the coding

sequences of annotated genes,” both “at [the] time of primary tumor sampling and at relapse” *Id.* at 57 (citing Ex. 1012, 66, 70) (emphasis omitted); *see also id.* at 58 (presenting additional arguments with respect to claims 1 and 5);

In opposing Ground 2, Patent Owner argues that Ley fails to address defects previously identified in Ground 1 (based on Leary alone). POR 36–37; Sur-reply 19–20. But as addressed above, Petitioner has demonstrated that claims 1–11, 13–19 are unpatentable in view of Leary alone. Patent Owner’s other arguments are premised on its constructions of “whole genome sequencing” and “capture probes.”

Patent Owner argues, for example, that the Ley authors would not have had access to sufficient resources to pursue whole genome sequencing at 90% sequence coverage, moreover, Ley’s method would not have been feasible without performing whole genome sequencing twice. POR 36–37; Sur-reply 19–20 (discussing the “data burden and cost constraints of WGS”). We do not find these arguments availing in light of our construction of “whole genome sequencing” as understood in light of the instant Specification. *See* Section II.C.1, above.

Patent Owner also appears to argue that Petitioner has not sufficiently explained how one of ordinary skill in the art would have “replac[ed] Leary’s PCR-based approach with capture probes” with a reasonable expectation of success. POR 37–38, 20. Again, we do not find these arguments availing in light of our construction of “capture probes” as understood in light of the instant Specification, and further in view of our determination that whole genome sequencing is described in Leary alone. *See* Section II.C.2, above. We also do not find Patent Owner’s remaining

arguments availing for the reasons set forth at pages 17–18 of Petitioner’s Reply and paragraphs 104–107 of Dr. Quackenbush’s Second Declaration, both of which we adopt.

Considering the record before us, and in light of our constructions of “whole genome sequencing” and “capture probes,” we find that Petitioner has shown persuasively that claims 1–11 and 13–19 are unpatentable in view of Leary and Ley.

F. Ground 3: Obviousness in View of Leary, Liao, and Chan

As to Ground 3, Petitioner challenges claims 1–19 as unpatentable for obviousness in view of Leary, Liao, and Chan. *See* Pet. 18, 58–77; Ex. 1020 ¶¶ 221–252; Ex. 1225 ¶¶ 104–107. Patent Owner opposes. POR 39–81; Sur-reply 20–26. Ex. 2031 ¶¶ 163–176. As Leary has been discussed above, we begin with an overview of Liao and Chan.

1. Overview of Liao (Exhibit 1009)

Liao discloses that “[m]assively parallel sequencing has recently been used in noninvasive prenatal diagnosis,” but notes that it is currently costly and sample throughput is still relatively low for a use as a molecular diagnostic. Ex. 1009, 92. According to Liao, “[r]ather than nonselectively sequencing the genome, target enrichment provides a logical approach for more efficient and cost-effective massively parallel sequencing because it increases the proportion of informative data from the targeted region(s).” *Id.* To obtain enriched DNA for sequencing, Liao used “an in-solution capture strategy” to obtain about 213-fold enrichment of fetal X chromosome DNA from maternal plasma samples. *Id.* at 92, 94 (disclosing sample set of 12 women carrying female fetuses and use of a “SureSelect Human X Chromosome Kit (Agilent)”), 98–100.

According to Liao:

[W]e incubated 500 ng of the amplified plasma DNA library with the capture probes for 24 h at 65 °C, in accordance with the manufacturer's instructions. After hybridization, we selected captured targets by pulling down the biotinylated probe/target hybrids with streptavidin-coated magnetic beads (Dynabeads M-280 Streptavidin; Invitrogen) and purified the targets with the MinElute PCR Purification Kit (Qiagen). Finally, we enriched the targeted-DNA libraries by 12-cycle PCR amplification with the SureSelect GA PCR primers (Agilent).

Id. at 94. Samples were sequenced on a Genome Analyzer IIx (Illumina) to identify fetus-specific SNPs. *Id.* Liao concludes:

After target enrichment in the present study, the mean sequence coverage increased 83-fold, which allowed 95.9% of the paternally inherited alleles within the targeted region to be detected. The high probability for detecting a fetus specific allele, if present, suggests that targeted sequencing would be a powerful method for accurately measuring the proportion of fetal DNA in a particular sample.

Id. at 100.

2. Overview of Chan (Exhibit 1008)

Chan discloses “shotgun massively parallel sequencing (MPS) of plasma DNA from cancer patients to scan a cancer genome noninvasively.” Ex. 1008, 211. According to Chan, “[t]he presence of tumor-derived DNA in the plasma of cancer patients offers exciting opportunities for the detection and monitoring of cancer.” *Id.* at 211. Chan states, however, that, “[m]ost of the previously published work on plasma DNA as a cancer marker has focused on the detection of specific and predetermined molecular targets known to be associated with cancer by means of such methods such as the PCR.” *Id.*; *see also id.* at 223 (noting that “solution-phase hybridization-based capture approaches have already been successfully used to enrich

plasma DNA for noninvasive prenatal diagnosis” (citing Liao, Ex. 1009)). In contrast to this targeted approach, Chan employed a “nontargeted random—or shotgun—sequencing approach” “to obtain a noninvasive, genomewide view of cancer associated copy number variations and mutations in DNA in plasma.” *Id.* at 212.

Illustrating the shotgun MPS approach, Chen follows the proportions of cancer-associated chromosomal aberrations in the plasma of four hepatocellular carcinoma patients over time, before and after resection. *Id.* at 211 (“Four hepatocellular carcinoma patients . . . were recruited. DNA was extracted from the tumor tissues, and the preoperative and postoperative plasma samples of these patients were analyzed with shotgun MPS.”), 212, 217–219; *see also id.* at 219–221 (MPS analysis on patient with breast and ovarian cancers). Chen concludes that shotgun MPS

would allow genomic aberrations harbored by tumor cells within a patient to be scanned at different levels of resolution, ranging from a global view of cancer-associated copy number aberrations to point mutations carried by different clones of tumor cells. This approach would allow qualitative and quantitative aspects of such aberrations to be followed serially, as evidenced by the changes we observed in the pre- and postsurgery plasma samples.

Id. at 221.

Chan further states that, although “the shotgun sequencing approach we have described is relatively expensive compared with many targeted approaches . . . cost is not expected to be as much of an obstacle in the near future.” *Id.* at 221. According to Chan,

[o]ne can also envision the possibility of using shotgun sequencing of a plasma sample upon presentation of a cancer patient, mining the shotgun sequencing data of plasma DNA for copy number aberrations and SNVs, and then specifically

targeting such sequences for serial monitoring purposes. In this regard, solution-phase hybridization-based capture approaches have already been successfully used to enrich plasma DNA for noninvasive prenatal diagnosis (34, 35). Hence, we see that the targeted and shotgun approaches can be used in combination for cancer detection and monitoring.

Id.

3. Overview of Petitioner's Argument

According to Petitioner,

Leary discloses performing an initial whole-genome sequencing step to identify mutations (polymorphisms) of interest in a tumor sample, and then interrogating those mutations serially by detecting them in the patient's blood at subsequent time points. To detect the mutations in the blood, Leary discloses generating subsets of nucleic acids from blood samples by contacting them with PCR primer capture probes that hybridize to the nucleic acid sequences harboring the mutations, and renders obvious sequencing those subsets of nucleic acids in a sequencing assay.

Pet. 58–59 (citing Ex. 1020 ¶ 222) (emphasis removed). Petitioner argues that one of ordinary skill in the art

would have found it obvious to modify Leary's protocol by, following an initial whole-genome sequencing step, generating subsequent subsets of nucleic acid molecules by contacting blood or plasma samples with solution-phase hybridization probes that target (capture) the additional types of polymorphisms taught in Chan, including SNVs/SNPs and CNVs, followed by next-generation sequencing of those subsets of nucleic acids. A POSA would have found it obvious to use such techniques either as a replacement for Leary's PCR primer capture probes, or as an adjunct to them, given, for example, that solution-phase hybridization probes can be optimal for identifying and detecting mutations like SNPs.

Pet. 59 (citing Ex. 1020 ¶ 223) (emphasis removed). More particularly, Petitioner contends a person of ordinary skill in the art would have been

motivated, with a reasonable expectation of success, to employ “solution-phase hybridization probes [as taught by Chan] in Leary’s workflow to generate subsets of nucleic acids, followed by NGS to sequence those subsets to detect the presence or absence of a polymorphism of interest, as claimed in steps 1(c)-1(f).” *Id.* at 60 (citing Ex. 1020 ¶ 224) (emphasis removed). In this respect, Petitioner notes that “the use of solution-phase hybridization followed by NGS to serially track tumor-associated polymorphisms in the blood following an initial WGS step is expressly contemplated by Chan[,]” which employs “shotgun MPS to obtain a noninvasive, genomewide view of cancer-associated copy number variations and mutations in DNA in plasma.” *See id.* at 60–61 (citing Ex. 1020 ¶¶ 225–227; Ex. 1008, 212, 221–222).

Dr. Quackenbush describes Chan as disclosing the use of “solution-phase hybridization-based capture probes that target particular polymorphisms identified in an initial whole-genome sequencing assay to serially monitor and detect cancer in a patient’s plasma over time.” Ex. 1020 ¶ 228. According to Dr. Quackenbush, “Chan’s disclosures would have motivated a POSA to employ solution-phase hybridization probes, including solution-phase hybridization probes targeting additional types of polymorphisms, followed by NGS in Leary’s methods.” *Id.*; *see also* Ex. 1225 ¶ 100 (“The use of solution phase hybridization capture in the proposed composition occurs after whole-genome sequencing, and before the sequencing of the catch obtained by using capture probes designed with the benefit of the whole-genome sequencing data. *See* Ex1020, ¶¶236–241.”). Dr. Quackenbush explains, for example, that Chan motivates modifying claim step 1(c) to include “solution-phase hybridization probes

targeting polymorphisms such as SNPs and CNVs, in addition to the PCR primer capture probes targeting rearrangements taught by Leary, or as an alternative to such PCR primer capture probes,” because “this combination, which would provide an even greater ‘numerical advantage’ in the cancer-related polymorphisms to be monitored, thereby increasing the sensitivity of the overall assay.” Ex. 1020 ¶ 232 (citing Ex. 1008, 222–223); Ex. 1225 ¶ 100. “Thus, a POSA would have been motivated to add to the fundamental methodology set forth in Leary the analysis of the additional types of polymorphisms (e.g., SNVs/SNPs and CNVs) that Chan found to be present in greater numerical quantities in patients’ cancer samples.” Ex. 1020 ¶ 232; *see also id.* ¶¶ 233 (discussing motivation “to aid the detection and monitoring of a larger number of polymorphisms in parallel, and to adapt the capture technique to the types of mutations identified in the whole-genome sequencing step”), 234–235 (discussing motivation to combine Leary with Chan’s shotgun sequencing approach, which “provides a numerical advantage that has the potential for more sensitive, precise, and representative detection of tumor DNA in plasma” (citing Ex. 1008, 222–223), ¶ 236 (“A POSA would have been motivated to utilize [Chan’s] capture probes following the initial step of performing a whole-genome sequencing assay in Leary’s protocol as well, given the significant overlap in methodology, including to improve detection and monitoring of CNVs and SNVs identified via whole-genome sequencing, as well as to enable the serial monitoring of a larger number of targets by using multiplexed sequencing of the hybrid capture products using NGS.”).

With particular relevance to dependent claim 12 (“wherein said plurality of capture probes are conjugated to beads), Petitioner notes that

Chan and Liao (an earlier publication with authorship overlapping that of Chan) disclose “conjugating [a] plurality of capture probes to beads when the hybrid capture probes are bound to magnetic beads to select the catch.” Pet. 76–77 (citing, e.g., Ex. 1009, 94; Ex. 1020 ¶¶ 250–252). Petitioner further relies on Liao as emphasizing the prior success and applicability of solution-based hybridization capture approaches in detecting single nucleic acid polymorphisms (SNPs) in plasma samples. *See, e.g.*, Pet. 66 (citing Ex. 1020 ¶ 233), 71–77; Ex. 1020 ¶ 241 (overview of combining Liao). Discussing the asserted combination with respect to dependent claim 12 (further requiring a “plurality of capture probes . . . conjugated to beads”), for example, Dr. Quackenbush testifies that

Leary in view of Chan and Liao discloses conjugating the plurality of capture probes to beads when the hybrid capture probes are bound to magnetic beads to select the catch, as routine in the use of hybrid capture. Specifically, Liao explains that “[a]fter hybridization, we selected captured targets by pulling down the biotinylated probe/target hybrids with streptavidin-coated magnetic beads (Dynabeads M-280 Streptavidin; Invitrogen) and purified the targets with the MinElute PCR Purification Kit (Qiagen).” Ex. 1009 (Liao) at 94. A POSA thus would have understood that solution-phase hybridization probes will be conjugated to beads as part of the well-known enrichment process.

Ex. 1020 ¶ 251 (further citing Ex. 1005, 183 (Fig. 1)).

4. Whether Chan Qualifies as Prior Art

Petitioner bears the burden of persuasion, by a preponderance of the evidence, that the challenged claims are unpatentable. 35 U.S.C. § 316(e). Petitioner has proffered Chan, which presumptively constitutes prior art under 35 U.S.C § 102(a) because it was published in October of 2012, and before the January 17, 2013, filing of the ’828 Provisional Application, upon

which the '394 patent relies for priority. *See* Pet. 17 (citing Ex. 1020 ¶ 39); POR 39; Ex. 1001, code (60).

To remove Chan as prior art, Patent Owner undertakes the burden to show that the claimed invention was conceived before the effective date of Chan, followed by reasonably continuous diligence until constructive reduction to practice. POR 39–76 (citing, e.g., *ATI Techs. ULC v. Iancu*, 920 F.3d 1362, 1369–70 (Fed. Cir. 2019)); Sur-reply 24. “Demonstration of such priority requires documentary support, from which factual findings and inferences are drawn, in application of the rules and law of conception, reduction to practice, and diligence.” *ATI Techs* at 1369 (quoting *In re Steed*, 802 F.3d 1311, 1316 (Fed. Cir. 2015)). Conception, for example, “must be proved by corroborating evidence which shows that the inventor disclosed to others his completed thought expressed in such clear terms as to enable those skilled in the art to make the invention.” *Coleman v. Dines*, 754 F.2d 353, 359 (Fed. Cir. 1985) (internal quotation marks omitted). “The conception analysis necessarily turns on the inventor’s ability to describe [the] invention with particularity.” *Burroughs Wellcome Co. v. Barr Labs., Inc.*, 40 F.3d 1223, 1228 (Fed. Cir. 1994) (explaining that “[a]n idea is definite and permanent when the inventor has a specific, settled idea, a particular solution to the problem at hand, not just a general goal or research plan he hopes to pursue”). Until the inventor can do so, the inventor “cannot prove possession of the complete mental picture of the invention.” *Id.*

Patent Owner contends that Chan does not qualify as prior art because “the inventors conceived of the claimed invention by no later than May 8, 2012 and diligently worked towards its reduction to practice via filing the [’828] [P]rovisional [A]pplication on January 17, 2013,” which, if

established, would mean the claimed invention has a priority date of May 8, 2012—prior to the publication of Chan. POR 39.

With respect to conception, Patent Owner asserts that the claimed invention was developed through Personalis’ “Accuracy Program.” POR 40. Patent Owner relies on Mr. West’s testimony that, “[m]y co-inventors and I met many times between November 2011 and May 8, 2012 to discuss ideas for the Accuracy Program.” *See* POR 42; Ex. 2032 ¶ 57. In support, Mr. West points to an email (with attachments) dated January 6, 2012, as indicating that his team was working on “‘augment[ing] NGS [next-generation sequencing] with other technologies to determine key variants,’ and ‘conduct[ing] follow-up genetic & non-genetic testing to validate results.’” Ex. 2032 ¶¶ 57–59 (citing Ex. 2046, 1 (cover email), 3 (“slide 2”), 10 (“slide 9”), 14 (“slide 13”)). Further referencing Mr. West’s testimony, Patent Owner argues that “[o]n January 13, 2012, Mr. West planned to supplement whole genome sequencing using hybrid capture “oligo-based pullouts.” POR 43 (citing Ex. 2032 ¶¶ 53–56; Ex. 2109, 2 (“Targeted re-sequencing using array or oligo-based pullout”), 3 (“Exon re-sequencing by pullout . . . Additional re-sequencing by pullout, to capture regions outside the exons”); Ex. 2102, 2).

Patent Owner argues that “[o]n April 2, 2012, Mr. West described Personalis’ laboratory plans.” POR 43 (citing Ex. 2036; Ex. 2032 ¶ 61). “Inventors met and exchanged ideas on May 3 and 4, 2012.” *Id.* (citing Ex. 2037; Ex. 2038; Ex. 2039; Ex. 2040; Ex. 2032 ¶¶ 62–65). And “[o]n May 8, 2012, Mr. West presented his ‘CEO Overview’ slides to the Personalis Board of Directors.” *Id.* (citing, e.g., Ex. 2033 ¶¶ 20–26; Ex. 2112). Patent Owner relies on Mr. West’s “CEO Overview,” and in

particular Slide 5 (reproduced below), as evidence that the inventors had conceived of the claimed invention as of May 8, 2012. *See generally, id.* at 45–61; Sur-reply 23–24.

Overview of Personalis Accuracy Program

Characterize errors	Informatic Solutions	Lab, Content Solutions
<u>Analysis</u> Rates by variant type & Q-score Underlying mechanisms Medical interpretation impact Regulatory impact	<u>Advances in 3rd party alg's</u> Aligner for > 200bp reads New versions, current alg's Other alg's (e.g. lobSTR)	<u>Combine technologies</u> WGS + exome Longer reads and/or inserts Multi-vendor Arrays (genotypes & SV's) Electrophoretic sizing Sanger validation
<u>Methods</u> Comparison with literature Studies on limited regions Whole genome analysis: Single genome metrics Same person twice Family inheritance (MIE) Platform comparison Arrays ILMN / CGI / Ion T... Sanger capillary	<u>Advances in our alg's:</u> SV's : Zygosity, accuracy Combine SV's & SNP's Broader use of Breakseq	<u>Content-based semi-custom</u> Array (genotypes & SV's) Pullout
<u>Prioritize solutions</u> Bio-informatic approaches Laboratory approaches Content approaches	<u>New Personalis alg's</u> Alignment degeneracy Genome-aware alignment VNTR-specific	<u>Custom sample preps</u> Problematic motifs Degenerate regions Phasing
	<u>Advances in reference</u> Ethnic major-allele ref's Iterative Admixed InDel's / SV's	<u>Content</u> Gold standard genomes Known healthy genomes
	<u>Flag or Filter Out</u> For problems not yet solved	

According to Patent Owner, Slide 5 “concisely describes the Accuracy Program.” *Id.* at 44 (citing Ex. 2112, 5).

Petitioner, however, challenges the credibility of Personalis’ invention story. *See generally*, Tr. 10:19–13:18. As an initial matter, Petitioner notes that “[n]amed co-inventor Bartha started work at Personalis on May 1, 2012—just a few days before its *latest* proffered bound for conception.” Reply 20. According to Petitioner, “[i]t is highly implausible that Dr. Bartha participated in conception within just a few days of starting at Personalis, particularly given that Mr. West’s May 8 Board of Director[’]s presentation, which is the basis for Personalis’s ‘no later than’ date, was drafted prior to May 8, . . . and Dr. Bartha is not listed as a participant at the May 3 and

May 4 meetings Mr. West discusses.” Reply 20 (citing Ex. 1222, 51:18–24; Ex. 2038–2040; Ex. 2032 ¶¶ 63–65).

In response, Patent Owner points to Mr. West’s testimony that upon joining Personalis, Dr. Bartha “was immediately tasked with working on the Accuracy Program under Mark Pratt.” Sur-reply 21 (citing Ex. 2032 ¶ 34). But other than at this high level of generality, Patent Owner provides no evidence of how Dr. Bartha allegedly contributed to any claim of the challenged patent.³³

Also pertinent to credibility, Patent Owner’s reliance on the May 8, 2012 “CEO Overview” is somewhat at odds with Mr. West’s implication that conception was at least substantially complete more than a month before Dr. Bartha’s start date. In particular, Mr. West testified that

all the pieces that came together to form the basis for the ’033 and ’394 patent, probably the last of those elements to come together was from a paper that was -- I believe it’s an exhibit here – that was published by Mike Snyder’s group at Stanford. The publication came out in March of 2012, and our group had seen a preview copy of it at least a month earlier. We worked very closely with Mike since he was a cofounder of the company.

³³ With respect to Petitioner’s related argument that co-inventor Luo was not hired until June 18, 2012, Patent Owner argues that Ms. Luo is only an inventor with respect to dependent claim 3. Reply 20; Sur-reply 21–22; Tr. 10:24–11:7, 12:3–11, 29:2–30:4. Although we recognize Petitioner’s argument that Ms. Luo only contributed to reduction to practice of claim 3, not conception, because Patent Owner’s antedation proofs address only the May 8, 2012 conception date for independent claim 1, for the purpose of this Opinion, it is sufficient that we apply our conclusion regarding the lack of conception of claim 1 to claim 3 and the other dependent claims.

Ex. 1222, 48:19–49:3. And although Mr. West later stated that “the last of the elements that, then, came together in the material that we presented to our board on May 8th,” the record does not explain what “last of the elements” pertinent to conception were added in the CEO Overview, nor what contribution Dr. Bartha made to the claimed invention. *See id.* at 49:12–14.

Relying on the testimony of Dr. Quackenbush, Petitioner next contends that the Accuracy Program cited as the basis for Personalis’ conception, was not a program to develop a minimum residual disease test such as the NeXT Personal MRD assay Patent Owner asserts is covered by the claims, but “a program to improve sequencing accuracy.” Reply 21 (citing, e.g., Ex. 1225 ¶¶ 116–125); POR 81. “Hence,” Petitioner contends, “Personalis’s first product – prototypes of which the patent expressly states were the source of its data – was its Exome Plus/Ace Exome product, released in February 2013,” whereas “Personalis’s NeXT Personal MRD diagnostic test was not launched until 2021,” nine years after Mr. West’s 2012 presentation. Reply 21 (citing Ex 2032 ¶38; Ex. 1001, 74:60-64); Ex. 1225 ¶ 119. While not dispositive, this testimony supports Petitioner’s contention that work related to the Accuracy Program was not directed to conception of the invention claimed in the ’394 patent.

Further supporting Petitioner’s contention that Patent Owner’s documents do not evidence conception of the claimed invention, Dr. Quackenbush credibly testifies that one of ordinary skill in the art would understand references to, for example, “characteriz[ing] errors” and performing “whole genome sequencing” on the “same person twice,” and “Pullout” in Slide 5, as well as “Repeatability,” and “Same person twice (or

more?), trying to keep everything else constant,” in Exhibit 2046, as directed to improving sequencing accuracy rather than to a program to develop a diagnostic assay relating to the inventions described in the challenged method claims. *See* Ex. 1225 ¶¶ 119–125 (citations omitted); Reply 21–22.³⁴ Accordingly, Dr. Quackenbush’s testimony weighs against a finding that Patent Owner’s evidence demonstrates a conception of the invention claimed.

Consistent with Exhibit 2046’s focus on sequencing accuracy, Petitioner argues “[t]here is no corroboration that, as of May 8, 2012, Personalis’s co-inventors conceived of a diagnostic assay involving the use of capture probes as claimed in step 1(c).” Reply 22–23 (citing, e.g., Ex. 1225 ¶ 125–126). In this respect we note Mr. West’s testimony that he understood “pullout” in Slide 5, as a reference to Agilent SureSelect biotinylated probe system. *See* Ex. 1222, 83:8–85:22, 88:7–16. We, nevertheless, find persuasive Dr. Quackenbush’s testimony that “a POSA reviewing Slide 5 in light of the overall record on conception would have insufficient information to determine how, specifically, Personalis intended to use the “Pullout,” other than that their purpose was to provide a laboratory solution to improve the accuracy of 2012 sequencing technology in some way.” Ex. 1225 ¶ 125.

Considering the evidence of record, we credit Dr. Quackenbush’s testimony that a person of ordinary skill in the art reading Personalis’ conception exhibits, including Exhibit 2112, “would have found, at most, an

³⁴ Dr. Furneaux offers no independent opinion on how one of ordinary skill in the art would have understood Patent Owner’s antedation proofs, or whether Chan is prior art. *See* Ex. 2031 ¶ 164.

incipient research project for improving the accuracy of sequencing technology that would involve the use of an undefined combination of various conventional techniques.” Ex. 1225 ¶ 128. We find persuasive, for example, Dr. Quackenbush’s testimony that Patent Owner provides no corroboration for an earlier conception of steps 1(d)-(g), which require assaying samples obtained from the same individual obtained *at different time points*. *See id.* ¶¶ 120–121, 126–127; Reply 24; Sur-Reply 23–24 (citing Ex. 2032 ¶¶ 249, 252–254, 258; Ex. 2109, 4 (slide 3); Ex. 2112, 3, 5, 13; Ex. 2036). In sum, we find that Personalis’ exhibits and related testimony do not evidence sufficient conception of the invention actually described in the challenged claims. *See, e.g.*, Ex. 1225 ¶¶ 120–128 (Dr. Quackenbush’s testimony about missing aspects of the claims in Patent Owner’s proofs, such that the “completed thought” is not shown).

Considering the above, we find that Patent Owner has not satisfied the burden of establishing that its claimed invention is entitled to an earlier priority date than Chan.³⁵

5. Obviousness in View of Leary, Chan, and Liao

Patent Owner argues that Ground 3 fails because the combination of Leary, Chan, and Liao fail to render obvious limitations 1(b), 1(c), 1(e), and 1(f). POR 76–81; Sur-reply 25–26; Ex. 2031 ¶¶ 165–175. Patent Owner’s arguments are not persuasive insofar as we find all these challenged limitations are satisfied by Leary alone.

³⁵ For the purpose of this Decision, we need not address Petitioner’s further contentions that Patent Owner has not established conception of claim steps 1(d)–1(g), and that the filing of the ’828 Provisional Application does not qualify as a constructive reduction to practice of the challenged claims. *See* Reply 23–25.

Patent Owner, nevertheless, argues that the shotgun sequencing approach described by Chan does not constitute whole genome sequencing. POR 77–78. According to Patent Owner, Chan sequences DNA from plasma samples, which are enriched in sequences bound to nucleosomes, whereas portions of the genome not bound to nucleosomes are degraded and, “largely missing from the DNA circulating in plasma.” *Id.* at 77 (citing Ex. 2031 ¶ 167). We do not find this argument persuasive given that the art recognized Leary to disclose whole genome sequencing, yet Leary *also* sequenced DNA from plasma samples. *See* Section II.C.1.b, above. We similarly credit Petitioner’s argument that Newman’s “list of exemplary whole genome sequencing references,” includes Chan. *See* Reply 25 (citing Ex. 2031 ¶ 107; Ex. 2152, 7). Still further, we credit Petitioner’s argument that Chan also discloses performing initial whole genome sequencing on a tissue sample, rather than on plasma-derived DNA, and that a “POSA would have been motivated to use [Chan’s] sequencing protocol in instances where high depth of coverage was desirable, for example to identify SNPs for subsequent monitoring.” *Id.* at 25–26 (citing Ex. 1008, 218; Ex. 1225 ¶¶ 97–98). As such, we find that one of ordinary skill in the art would have understood Chan to disclose whole genome sequencing.

Patent Owner also contends that “Chan’s supposed use of solution-phase hybridization capture is conducted *before* the sequencing—not after,” as required by claims step 1(c). POR 79. But as the Petitioner points out, “the proposed combination performs an initial WGS step followed by targeted resequencing using capture probes; thus capture probes are used after WGS and before resequencing.” Reply 26–27 (citing Ex. 1225 ¶¶ 99–100). We find Patent Owner’s argument unavailing because the overall

structure of the challenged claims evidences no linkage between the WGS-related language of steps 1(a)-(b) and any other element. *See* Reply 23–24; Tr. 31:23–32:21. As aptly noted by Petitioner, the whole genome sequencing data resulting from claim steps 1(a)-(b) “could be immediately discarded and the claim could still be met.” Reply 23. Although Patent Owner’s counsel suggests that “a person of ordinary skill in the art would understand that the whole-genome sequencing from the limitation 1A does have a cascading effect on other claim limitations,” this attorney argument does not outweigh the plain language of claim 1. Tr. 32:6–12; *See Chef Am., Inc. v. Lamb-Weston, Inc.*, 358 F.3d 1371, 1374 (Fed. Cir. 2004) (“we construe the claim as written, not as the patentees wish they had written it”).

With respect to Patent Owner’s remaining arguments, we credit, and adopt, the argument and evidence set forth on pages 25–27 of Petitioner’s Reply and Paragraphs 97–100 of Dr. Quackenbush’s Second Declaration.

Considering the record before us, and in light of our constructions of “whole genome sequencing” and “capture probes,” we find that Petitioner has shown persuasively that claims 1–19 are unpatentable in view of Leary, Chan, and Liao.

III. CONCLUSION

Having analyzed the entirety of the record and assigning appropriate weight to the evidence cited by Petitioner and Patent Owner, and for the reasons we have set forth above, we determine that Petitioner has established, by a preponderance of the evidence, that claims 1–19 of the ’394 patent are unpatentable.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has shown by a preponderance of the evidence that claims 1–19 of the '394 patent are unpatentable;

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.³⁶

Claims	35 U.S.C. §	Reference(s) /Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–11, 13–19	103(a)	Leary	1–11, 13–19	
1–11, 13–19	103(a)	Leary, Ley	1–11, 13–19	
1–19	103(a)	Leary, Liao, Chan	1–19	
Overall Outcome			1–11	

³⁶ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

IPR2023-00224
Patent 11,384,394 B2

FOR PETITIONER:

Andrew E. Krause
IRELL & MANELLA LLP
akrause@irell.com

Amir Naini
NAINI P.C.
amir@nainipc.com

FOR PATENT OWNER:

Derek C. Walter
Adrian Percer
WEIL, GOTSHAL & MANGES LLP
derek.walter@weil.com
adrian.percer@weil.com